



TrustSystem 10.0--Electroacoustic Testing System

User Manual

RSTECH (BEIJING) co. LTD



End User License Agreement

The enclosed Software is provided to the End-user of the Software by Rstech, limited, for use only under the terms set forth in this Agreement. Rstech reserves and right not expressly granted to the End-User. The End-User owns the disk on which the Software is recorded, but Rstech retains ownership of all copies of the Software itself. The End-User assumes sole responsibility for the installation, use and results obtained from use of the Software.

1. License. Rstech grants to End-User a limited, non-exclusive and nontransferable license to install, maintain and use the Software in object code form on a single computer owned or leased by End-User solely in connection with the End-User's own business. End-User may make one copy of the computer on which the Software is installed. The Software is protected by copyright law. As an express condition of this License, the End-User must reproduce on the copy Rstech's copyright notice and any other proprietary legends on the original copy supplied by Rstech.
2. Restrictions. End-User agrees that the Software is a proprietary product and that all right, title and interest in and to the Software, including all associated intellectual property rights, are and shall at all times remain with Rstech.
 - (1) The End-User shall not copy the Software in whole or in part, other than is essential for the proper operation remain with Rstech.
 - (2) The End-User shall not modify, translate, reverse-engineer or decompile the Software except to the extent permitted by law.
 - (3) The End-User shall maintain the Software in confidence and ensure that it is protected from unauthorized copying or disclosure by measures that are no less stringent than those it uses to protect its own valuable information and that are, in any case, no less than reasonable in the circumstances.
 - (4) The End-User shall prohibit the use of the Software by anyone other than the End-User, its employees and agents.
 - (5) The acknowledgement of the rights in the Software shall not be removed from the Software or any installation of it.
 - (6) The End-User shall not transfer or assign the End-User Agreement except to an Affiliate of the End-User.
3. Protection and Security. End-User agrees that the Software contains trade secrets, proprietary information and copyright material of Rstech. End-User agrees to use its best efforts and to take all reasonable steps to safeguard the Software to ensure that no unauthorized person shall have access thereto and that no unauthorized copy,



publication, disclosure or distribution, in whole or in part, in any form, shall be made, end-User acknowledges that the Software contains valuable confidential information and that unauthorized.

4. Termination. This License is effective until terminated. This License will terminate immediately without notice from Rstech if the End-User fails to comply with any of its provisions. Upon termination the End-User must destroy the Software and all copies thereof. End-User may terminate this License at any time by destroying the Software and all copies thereof.
5. Enhancements. From time to time Rstech may, in its sole discretion, advise the End-User of updates, upgrades, enhancements or improvements to the Software and/or new releases of the Software, and may license the End-User to use such Enhancements upon payment of price as may be established by Rstech. All such Enhancements to the Software provided to the End-User shall also be governed by the terms of this License.
6. The validity construction and performance of this Agreement shall be governed by and interpreted in accordance with the laws of China.
7. Acknowledgment. By opening this package and/or installing this Software, the End-User acknowledges that it has read this license, understands it, and agrees to be bound by its terms and conditions. This software is protected by copyright and other patents and is provided under license from RSTECH Limited.



目录

1.	Introduction	5
2.	Installation and Start	5
2.1	Software Installation	5
2.2	USB Key Installation	6
2.3	Start	7
3.	Main Interface	7
3.1	Pull-down Menus	8
3.2	Test Control	10
3.3	Test Sequences/Project	11
4.	Setup	11
4.1	Hardware	11
4.1.2	Sound Card	12
4.1.3	Interface	12
4.1.4	NI DAQ	13
4.2	Calibration	13
4.2.1	Microphone Cal.	13
4.2.2	Amplifier Cal.	14
4.2.3	Mouth Cal.	15
4.2.4	Soundcard Cal.	15
4.3	Stimulus Editor	16
4.4	Tools	17
4.4.1	Signal generator	18
4.4.2	Level meter	18
4.4.3	Spectrum Analysis	18
4.4.4	Noise Level	19
5.	Test	20
5.1	Acquisition	20
5.2	Analysis	21
5.2.1	Frequency Response	22
5.2.2	Phase	23
5.2.3	Impedance	24
5.2.4	THD	24
5.2.5	Rub&Buzz	25
5.2.6	Spectrum	26
5.2.8	Current	26
5.2.9	RS Value	27
5.3	Processing	28
5.4	Message	38
5.5	Com Read-Write	39
5.6	Loop	39
5.7	Go to	40
5.8	Command	40
5.9	SN	41
5.10	Save	41
6.	Test Case	42
6.1	Microphone Test	42
6.2	Earphone Test	44



1. Introduction

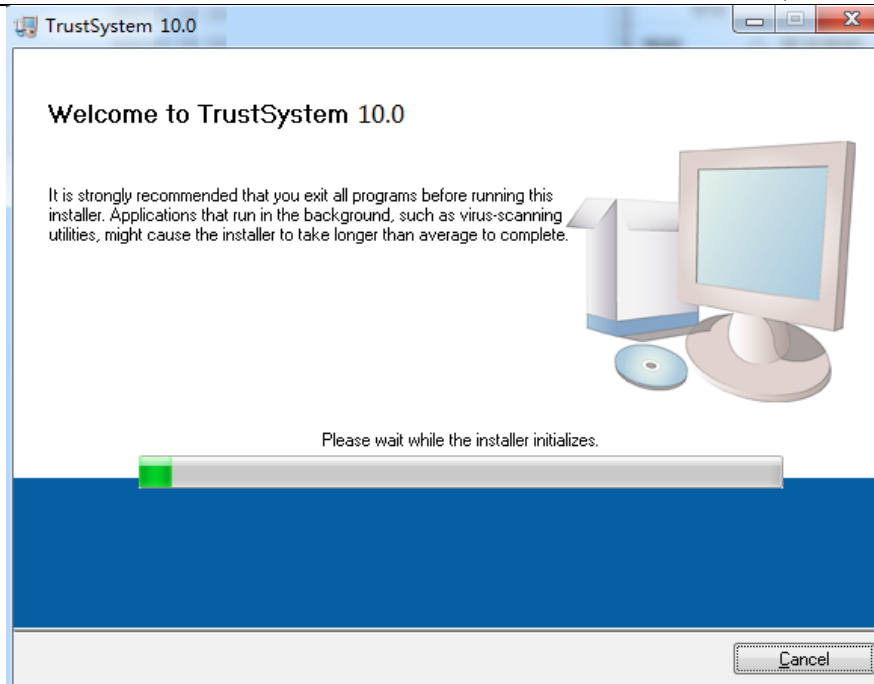
TrustSystem is one of the most convenient and easy-operating electroacoustic testing systems for the product line usage based on the personal computer. The System can test the cellphone & Tablet PC fast and accurately. TrustSystem is the System which you can trust. TrustSystem compute powerfully which could guarantee the precision and the speed of all parameters testing. TrustSystem integrates with different modules at different fields for various customers, can satisfy different projects, and test many products. It will upgrade constantly to meet the customer's demands, keep pace with times and engage in exploitation, innovation and perfection which would be the best choice for the product line.

Application		Test Items	
◇ Cellphone	◇ Tablet PC	◇ FR	◇ THD
◇ Microphone	◇ PC	◇ Impedance	◇ Polarity
◇ Bluetooth Headset	◇ Headset	◇ Phase	◇ Cross Talking
◇ Audiphone	◇ VOIP Phone	◇ T-SParameters	◇ SNR
◇ Smartwatch	◇ Speaker	◇ Rub & Buzz	◇ Loudness
◇ Intelligent speaker	◇ Receiver	◇ Electric Power	◇ Sound Power
◇ VR Glasses	◇ UAV	◇ Crosstalk attenuation	
◇ ANC Headset			

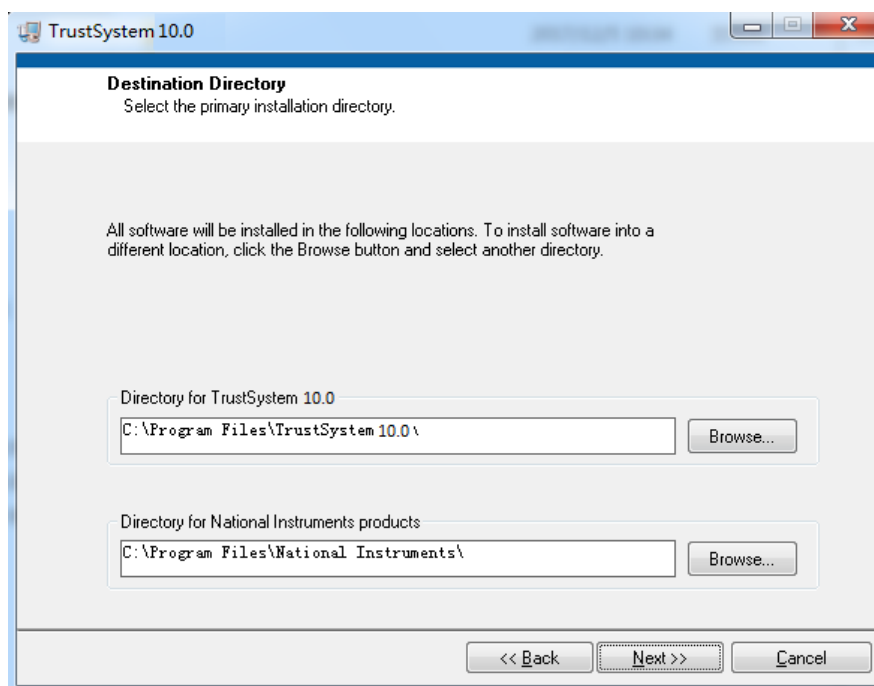
2. Installation and Start

2.1 Software Installation

Place the TrustSystem10.0 installation CD in the CD-ROM drive of your computer. You can begin the software installation by double-clicking  **setup.exe** on the file Installer.



Set the installation path, and click Next to continue the installation.



The shortcut will be generated on the desktop while the installation is complete.

2.2 USB Key Installation

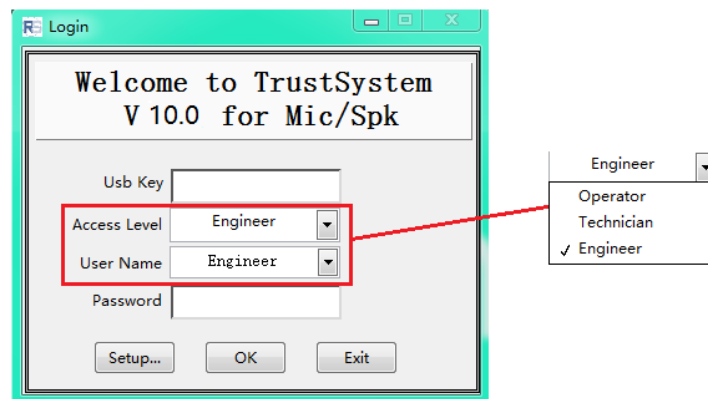
Insert the USB key to your computer, it will be installed the drivers automatically, and then replace the file  **Key.rs** in the software installation directory, the key driver are finished when the light in the USB key are lighting.



2.3 Start

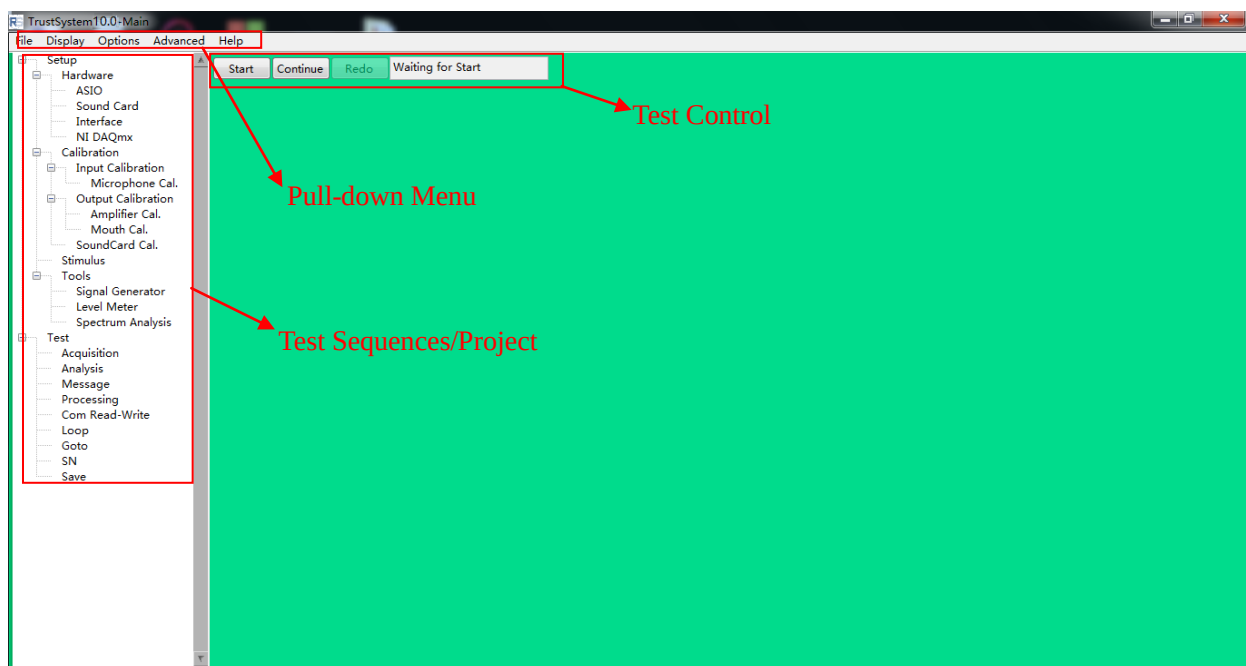
To view and change the system login settings, select Re-login from the File pull-down menu on the main TrustSystem menu bar. The Login screen also appears when first starting the TrustSystem program. Login allows the TrustSystem Engineer (3 levels: engineer, technician and operator) to prevent use of certain test system functions by other users. For instance, Calibration, and Setups can be password protected from unauthorized users.

The Engineer level is the highest access level, and therefore all functionality is accessible. The Technician level allows access to Calibration procedures, and may measure, print and save data. The Operator level can only measure and print.



3. Main Interface

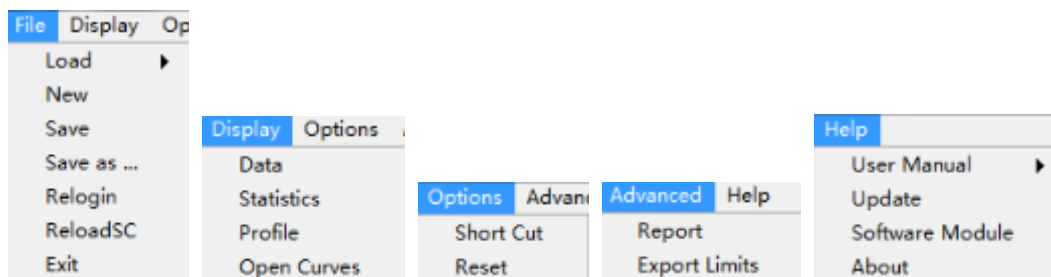
TrustSystem main interface is shown below, mainly including Pull-down Menus, Test Sequences/Project and Test Control.





3.1 Pull-down Menus

To view the selection in a pull-down menu, left click on a heading menu and click the desired selection.



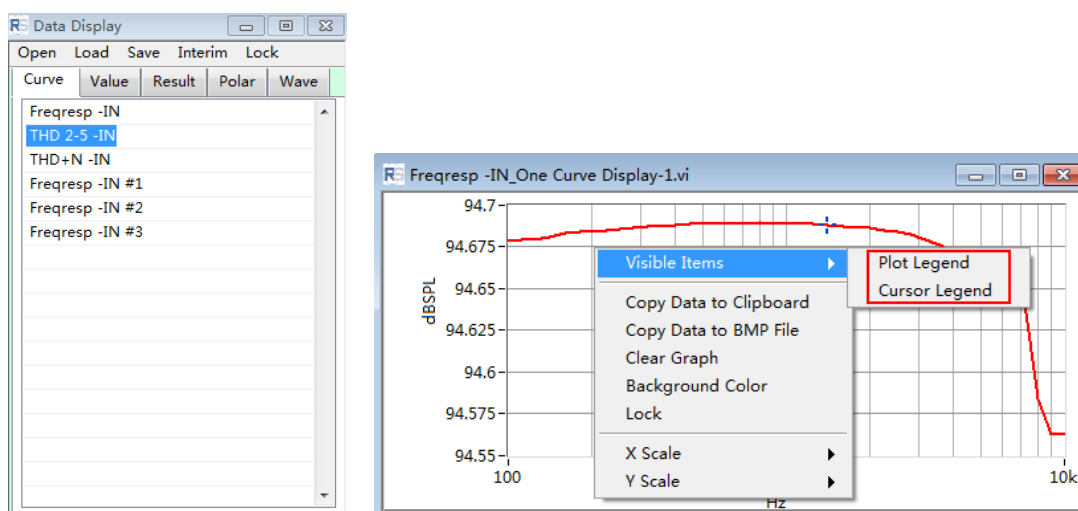
- File

When all the settings have done, you can save the parameters as a project, select the **save** or **save as** from the **File** pull-down menu bar, or you can load the project, when you want to relogin the testing system, you can choose **relogin** from the **File**, when you done this, TrustSystem load the last settings.

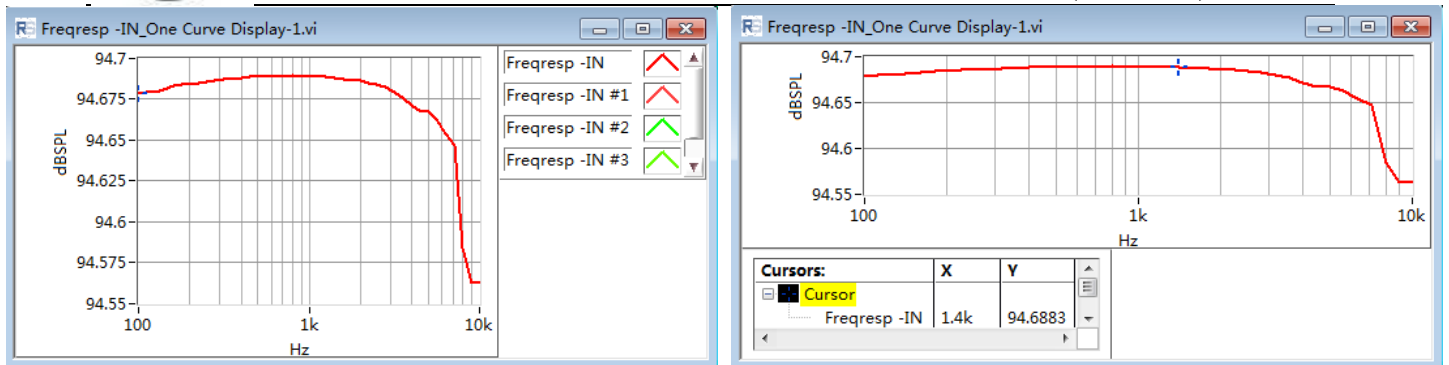
- Display

The display contains 3 items: Data, Statistics and Profile. To know the testing result, please pull-down the main menu **Display**, three parts are included:

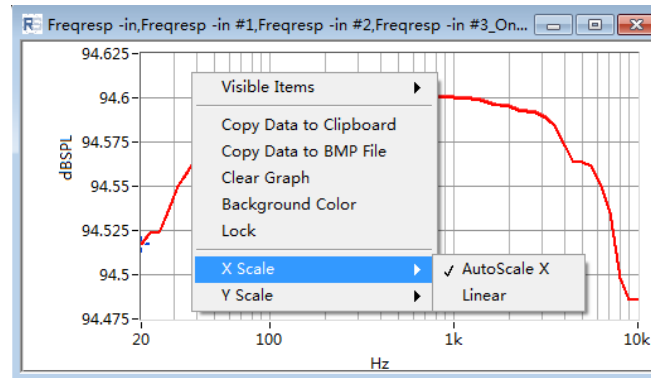
- Data: Click the Curve to check the curves and values that you want to display; click Value to check the values; click Result to check the limited curves and values that you want to display; click Polar to check the polarization; click Interim to check the current curve temporarity; click Load to check the data you have saved.



Select the curve and click Open to view the curve data; You can also drag the data directly to the display box, and right-click on Visible Items to add the illustrations and cursor.



You can select X/Y scale and cancel AutoScale to set the fixed ratio of X/Y axis.



Polar and Wave enable you to view directed graph and the time domain signals sent/received.

- Statistics: to display all the curves with limits and the single point value defective rate.

Curve/Value	Pass	All	%
Freqresp -IN	3	3	100.00
THD 2-5 -IN	3	3	100.00
THD+N -IN	3	3	100.00
All	3	3	100.00

- Profile: to check the run Project, access level, S/N and Time

Lot: MIC
SN L: 7200051082
☒ SN R: 12
Test: 1
Time(s):

● Options

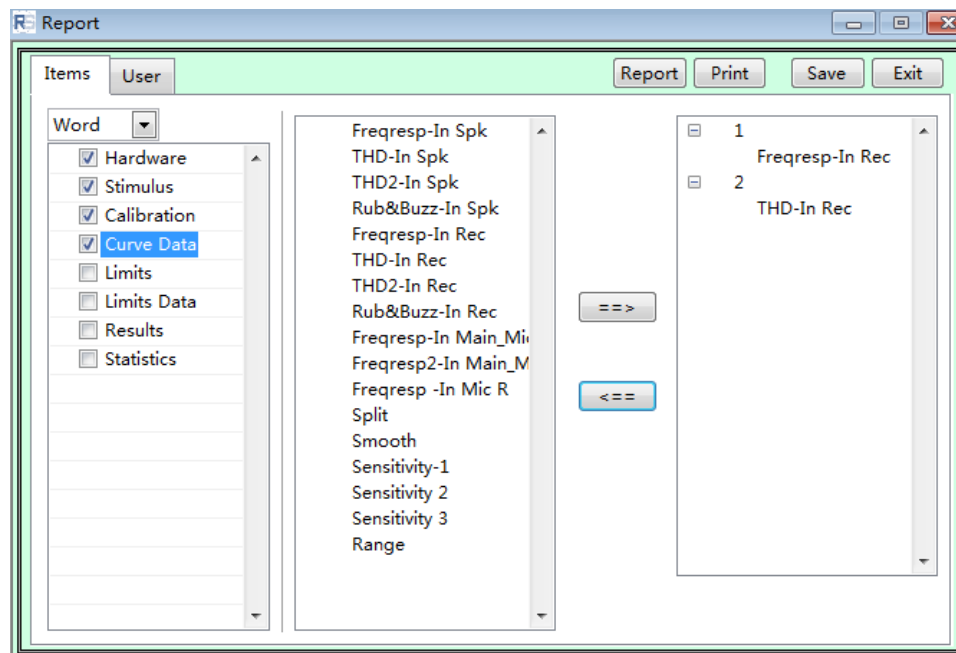
If your software broke down, select the Reset from the main menu Options, can reset your system. Short Cut is used to set the shortcut keys; If the current Project is confusing, reset



the program by using Reset.

- Advanced

You can report the testing information via clicking reports button, printing to Microsoft word, or print it by printer. Selecting the item to print, it's the testing conditions and the testing information, checking the curve data and Limit Data, it will print testing graph and the limits. Your company name can be included on your reports, check the company name item, entering your company name, of cause you can print the operator name. When all the settings is done, click Reports to create a word file, or click the Print to print the file directly through the Printer.

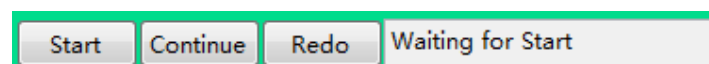


- Help

To know more the information about the software, select the pull down main menu Help, you can select the mainul, versions, software module ect.

3.2 Test Control

Test Control allows you to click Start/Stop to control the beginning/end of the test process, or click Continue to conduct continuous testing.



Note: Click the Redo button to retest, and the test results will automatically override the last test result (the Redo button can only be activated by adding the Save item to the test sequence).

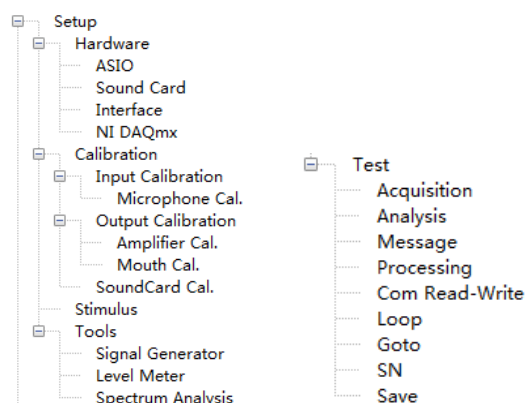


3.3 Test Sequences/Project

You can build your testing project in the left window which is tree-formed structure. Basically it is made up of two parts: Setup and Test.

Setup part can define the hardware and stimulus signal, and do the calibration, the tools such as generator, level meter and Spectrum Analysis are also integrated with.

Test part contains the flexible components and useful module to program your test project. There are mainly nine segments including acquisition, signal analysis and processing, communication port order, message, curve display, test result, data save, and report.

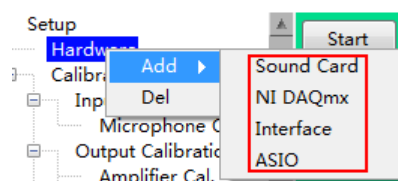


4.Setup

Before the test, you need to define the hardware, do the calibration, set the stimulus signal and so on.

4.1 Hardware

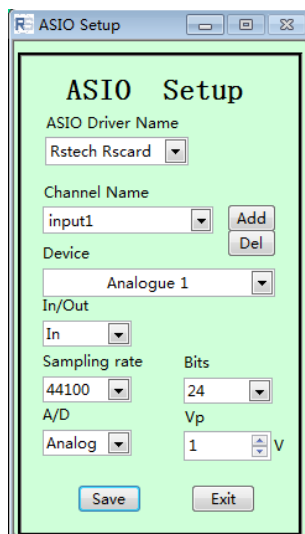
To view and change the system's soundcard settings, select Hardware from the right clicking of Setup, right click to add the soundcard. TrustSystem can work with a variety of soundcards, including Sound Card (all the device identified as a sound card by PC), NI DAQmx (NI collection card series), Interface (sound card with serial port), ASIO (Multi-Channel Combined Acoustic System developed by Rstech). There are a wide variety of sound cards available with varying degrees of performance but we recommend that you use the one of the certified sound cards from Rstech.



4.1.1 ASIO Setup



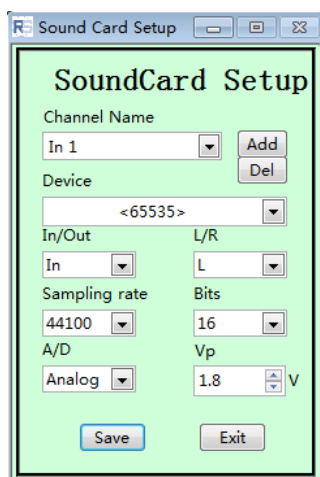
The ASIO Setup interface is mainly used for setting the parameters of Keyconnect (Multi-Channel Combined Acoustic System developed by Rstech).



- Driver Name
- Channel name
- Device
- In/Out channel
- Sampling rate
- Bits
- A/D
- Vp

4.1.2 Sound Card

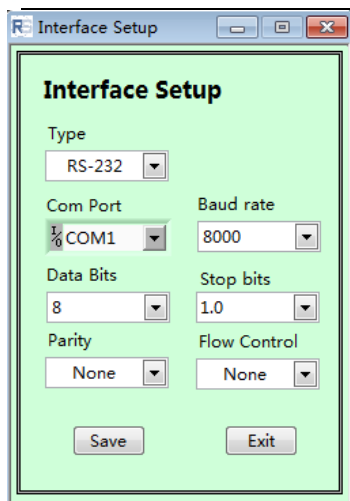
The Hardware Step has a multi-channel capability of TrustSystem: Inputs and Outputs are defined separately; Channels are defined only one at a time, not in pairs; Channel can be added, deleted or renamed.



- Channel name
- Device name
- In /Out
- Channel L/R
- Sampling Rate
- Bit Depth
- A/D
- Vp

4.1.3 Interface

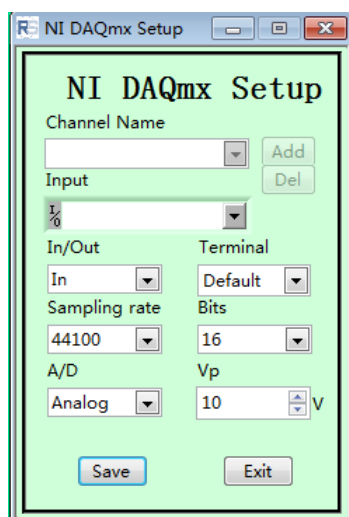
To set the communication port in the foot switch test, the following parameters in the Interface Setup could be set.



- Type: serial port type selection, Footsw in default
- Com port: communication port selection
- Baud rate: the default value is 8000
- Data Bits: the default value is 8
- Stop bits: the default value is 1.0
- Parity and Flow control: the default value is None

4.1.4 NI DAQ

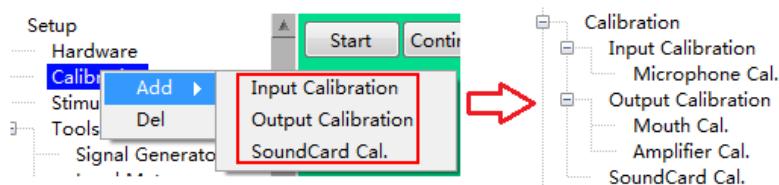
In the NI DAQmx Setup interface, you can set the following parameters for NI collection card series:



- Channel name
- Input/Output channel
- Terminal
- Sampling rate
- Bits
- A/D
- Vp

4.2 Calibration

Right click “Calibration” from Setup tree to add the calibration items. TrustSystem10.0 supports calibration of microphones, artificial mouths, power amplifier and sound card.

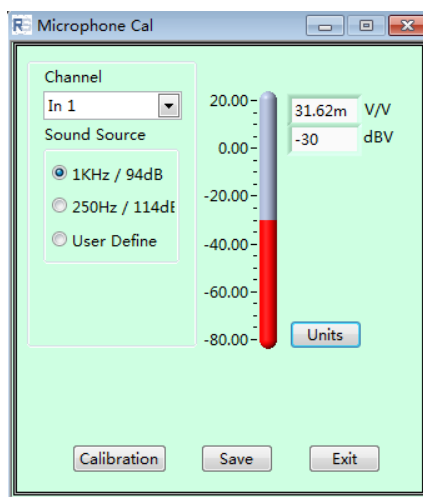


4.2.1 Microphone Cal.

The Microphone Cal is used to calibrate the complete TrustSystem test setup including

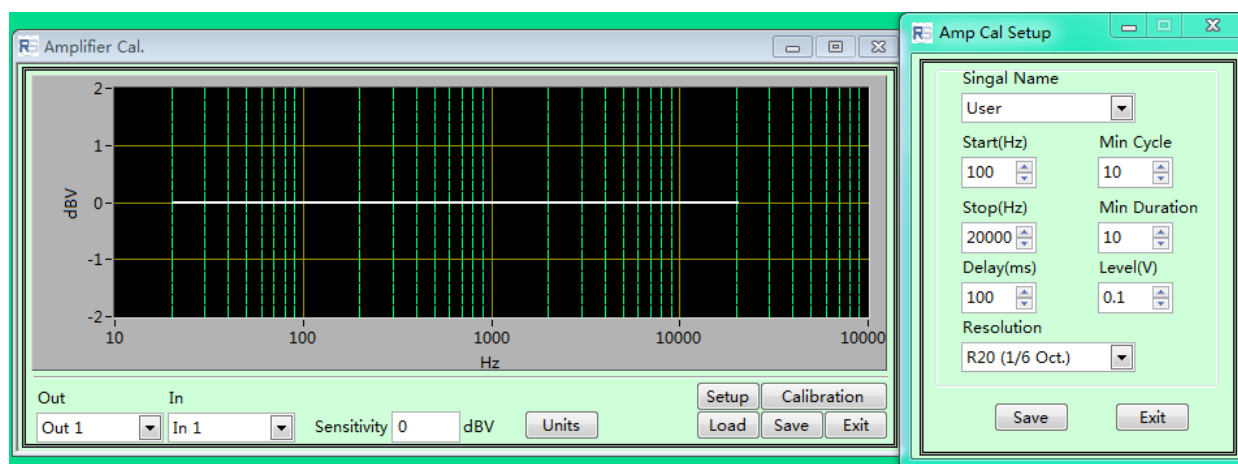


transducers (e.g. microphones and sound sources). This allows absolute measurements of acoustic, electroacoustic, electrical, and electronic devices. The step calibrates the entire system. The active Project's Calibration Step input sensitivities are also used for calibrating the virtual instruments accessed in calibration menu. Choose the input channel that microphone connecting to the soundcard, and sound sources, calibration is ok when the value becomes stabilization, choose the average button on the display menu.



4.2.2 Amplifier Cal.

To view and change the system's amplifier gain, select amplifier from the calibration menu on the test sequence menu. The amplifier setting should be used when there is an amplifier between the output of the sound card and the device under test. This might be required for loudspeaker measurements. To drive different loads. Enter the gain of your amplifier or run the calibration test measure it. If you are going to measure it, make sure your input is not calibrated first. Calibration parameters can be set in the setup button, the frequency range and the octave can be set by your demands, the level and the delay uses the default values(recommend).





4.2.3 Mouth Cal.

Mouth should be calibrated when you want to use an acoustic mouth simulator or sound source for testing microphones close to the source with a swept sine at a constant sound pressure level:

Step 1: using a calibrated reference microphone, place the microphone in the same position that you intend to measure your microphone or hearing aid under test, and select optimum channel.

Step 2: click Mouth Cal in the calibration area to begin the calibration procedure for your mouth simulator.

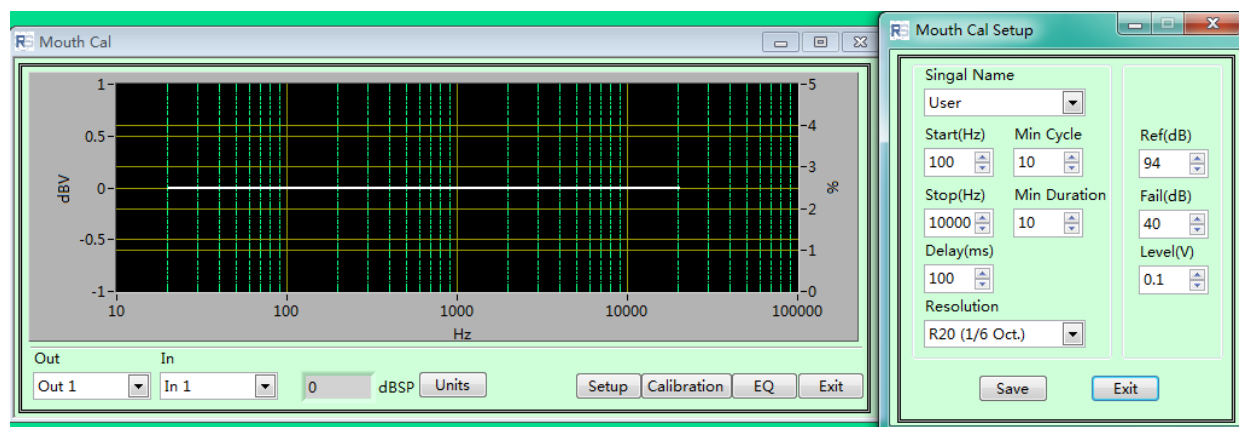
Step 3: click Setup to set the parameters about calibration stimulus, start/stop frequency, delays, and the reference SPL (generally is 94dB), and the fail limits (the limits for the mouth's frequency response), select the OCT and the level.

Step 4: click the calibration, the status of EQ SPL becomes about 94dB, click EQ button.

Step 5: click EQ button, the frequency response will be test.

Step 6: after clicking Continue, TrustSystem will generate an equalized stimulus and play it through the Mouth. The typical equalized response is ± 0.2 dB from 100Hz to 10kHz.

Step 7: save it, then TrustSystem will update the mouth sensitivity, and will equalization the mouth's Frequency response curve.



4.2.4 Soundcard Cal.

The process of soundcard calibration is as follows:

Step 1: short joint the input channel and the output channel of soundcard;

Step 2: set the Vp value of the input/output channel to 1V;

Step 3: select the input/output channel in Soundcard Cal screen; click "Setup" to set the start/stop frequency of the calibration, the signal voltage and the 1/n OCT, then click to save and

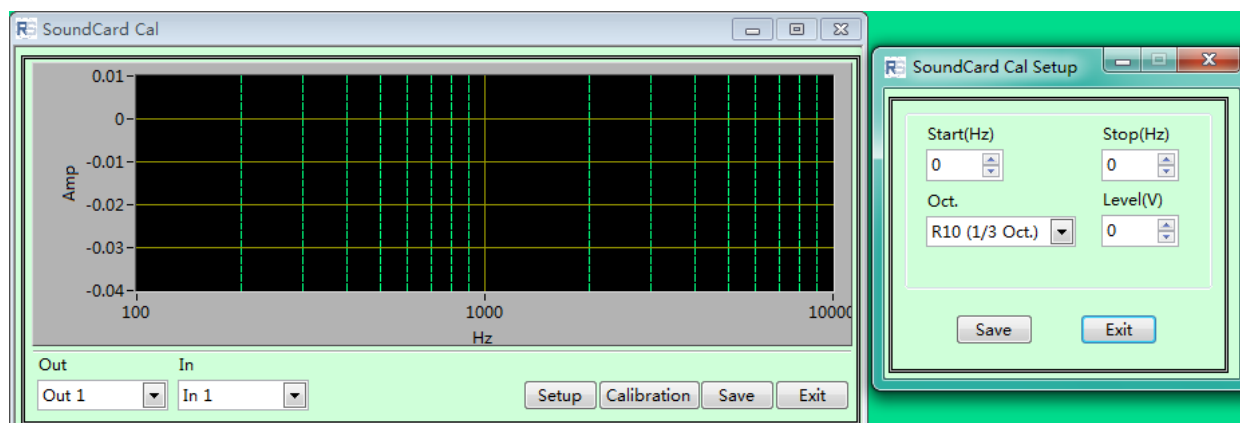


exit;

Step 4: click Calibration and measure the voltage value of the output channel according to the prompt;

Step 5: reconnect the input and output channels, and fill in the measured output terminal voltage; click “OK” to calculate the Vp value of the channel;

Step 6: click to save the Vp values and then exit.



4.3 Stimulus Editor

To view and change the system's Stimulus settings, select Stimulus from the Setup menu and add the stimulus. The Stimulus Editor will appear indicating the current test signal settings for the selected test.

- 1) **Output Channel:** select the right channel you setting in Hardware, and match the signal output that you connect to the soundcard.
- 2) **Signal Name:** enter the name of the signal that you setting.
- 3) The available **stimulus** including:

Frequency stepped-sine sweep: a stepped-sine sweep use an integer number of cycles at each frequency step, ensuring a smooth transition from frequency to frequency. This ensures significantly less transducer setting time and results in faster and more accurate measurements.

Log Amplitude Sweep at s single frequency.

Noise: featuring Pink and White Noise, with scalable Duration(s) and Band limits.

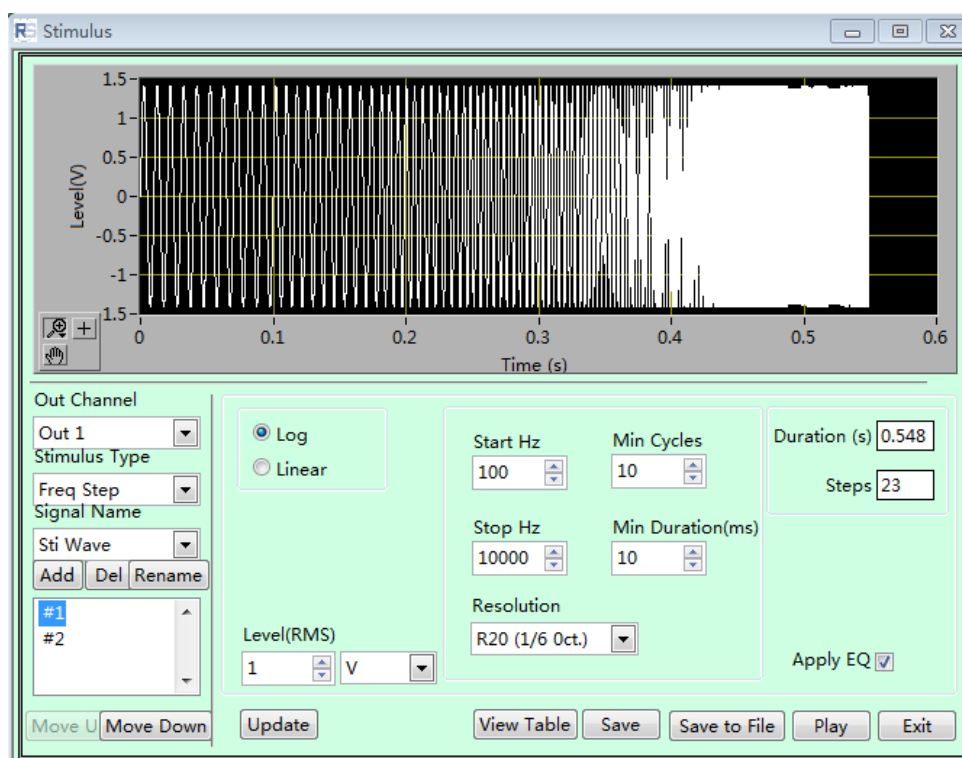
User Define: user can define different frequency point to sweep.

- 4) **Resolution** determines the number of measurable frequencies, including the start and stop frequency. You may choose standard ISO frequency steps such as R10 and R20, which corresponds to preferred 1/3 and 1/6 octave center frequency steps.
- 5) **Level:** enter the correct test level in either dB or linear units with keybosrd or the



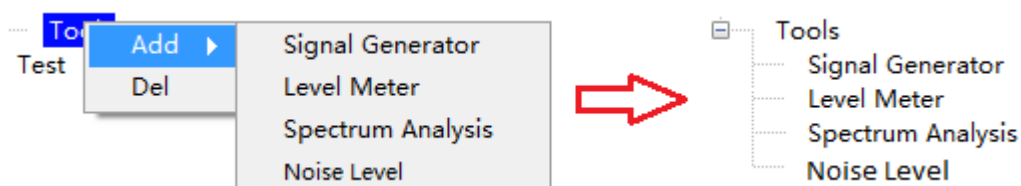
mouse. The output level and units are influenced by the output calibration sensitivity and units. The level set is the root mean square (RMS) value at the terminals of the device under test.

- 6) **Cycles & Duration:** To ensure proper measurement accuracy, each Sine step must have minimum number of cycles and a minimum duration. To properly measure the level of sine, you need at least ten cycles. In the presence of background noise, it is likely more cycles will be required. Because of noise, you also need a minimum duration to achieve a precise measurement. Min Cycles is the minimum number of cycles of sine at each step. Min Duration is the minimum dwell time, in milliseconds, at each step.



4.4 Tools

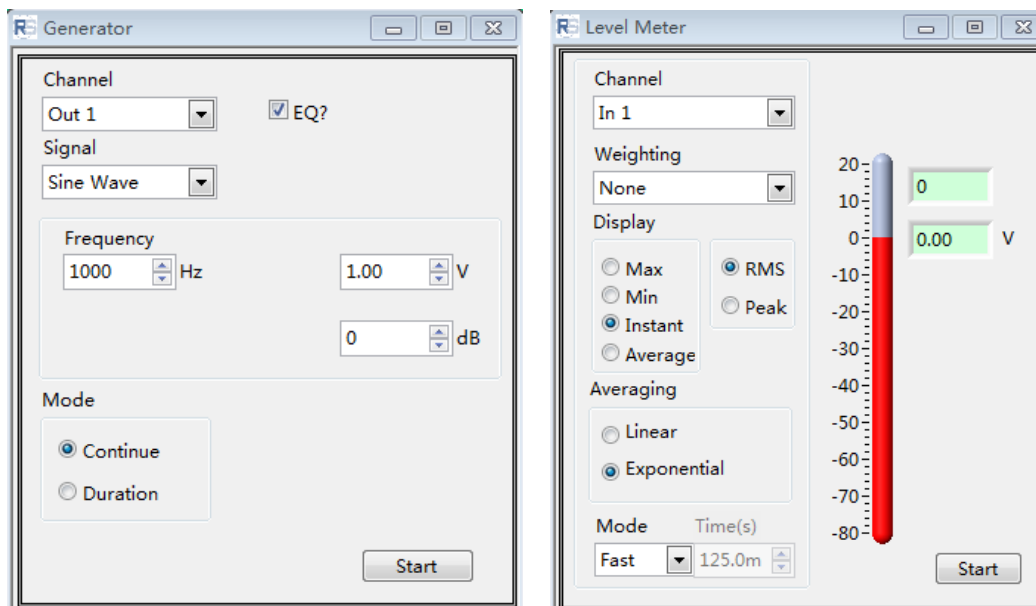
To view and change the system's tools, right click the setup, and add the tools, the same way to add the specify tools, including: signal generator, level meter, spectrum analysis and noise level.





4.4.1 Signal generator

The Signal Generator can be used for specify frequencies and output levels, you can play sine wave (noise or wave file) continuous or for a specified duration of time.

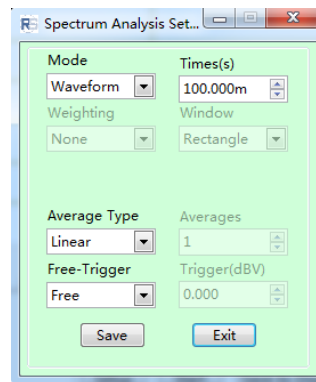
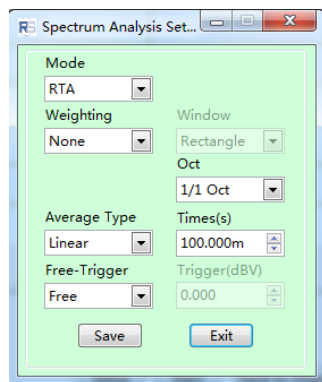
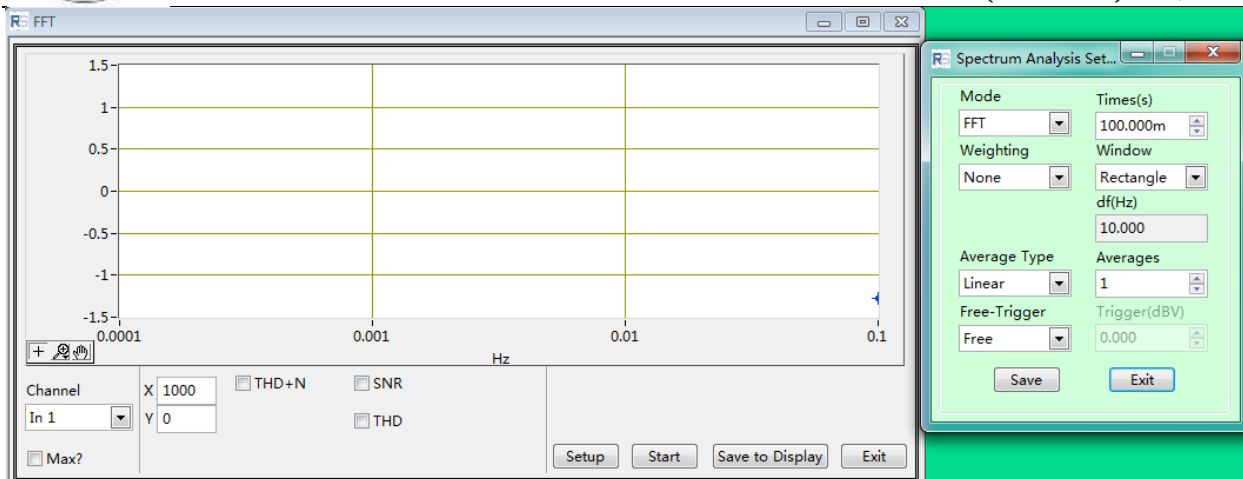


4.4.2 Level meter

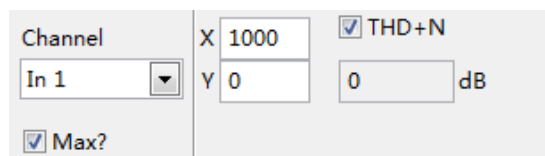
The Level meter can be used for measuring overall signals, choose between linear and exponential averaging and the durations of averages you wish to make. If you want the level meter to run for a preset time, choose Lin and the number of averages multiplied by the record time will determine the total measurement time. If you choose Exp, the level meter will run continuously until click OK or Exit in the level meter window.

4.4.3 Spectrum Analysis

Spectrum Analysis (FFT) can be used for analyzing the noise from the DUT, you can set up the frequency range to analysis the signal, can be added weighting to the signal from the DUT, as: A-weighting, C-weighting or B-weighting. Can analyze the RTA (real time analyzer) use the FFT at different octave.



In addition, you can check HD, THD+N, SNR to calculate the signal distortion results during the test, check Max to show the maximum value of the test in real time.

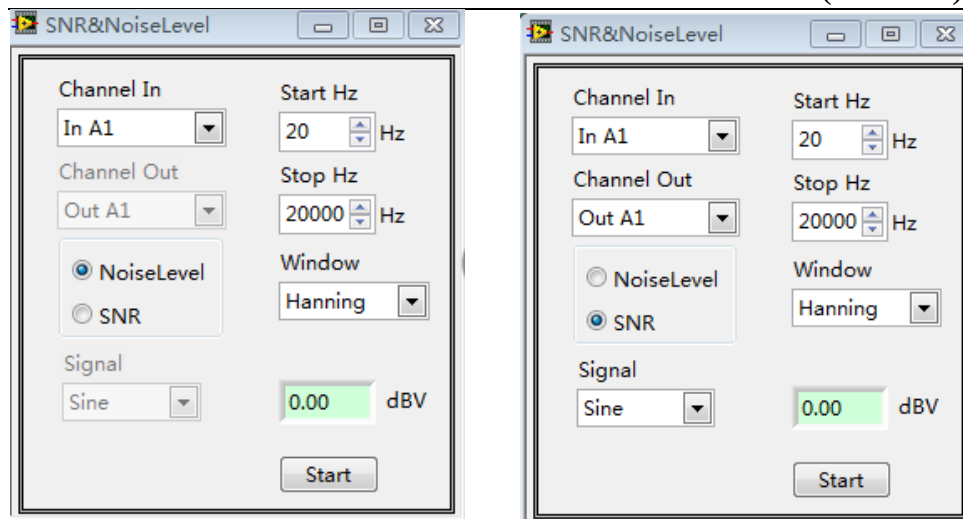


4.4.4 Noise Level

Noise Level can be used to compute the SNR (Signal to Noise Ratio) and background noise (Noise Level). The main steps is as follows:

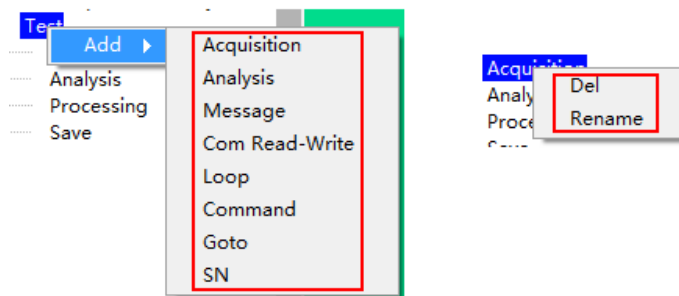
Noise Level: 1) select the Input channel; 2) set the start/stop frequency; 3) add the window function; 4) click Start button to calculate the SNR, and the value real-time display in the box **0.00 dBV**.

SNR: 1) select the Input / Output channel; 2) select the signal type (**Sine**, use the sine signal to test SNR in loop; **Wave File**, use a special composite signal to calculate the SNR); 3) set the start/stop frequency; 4) add the window function; 5) click Start button to calculate the Noise Level, and the value real-time display in the box **0.00 dBV**.



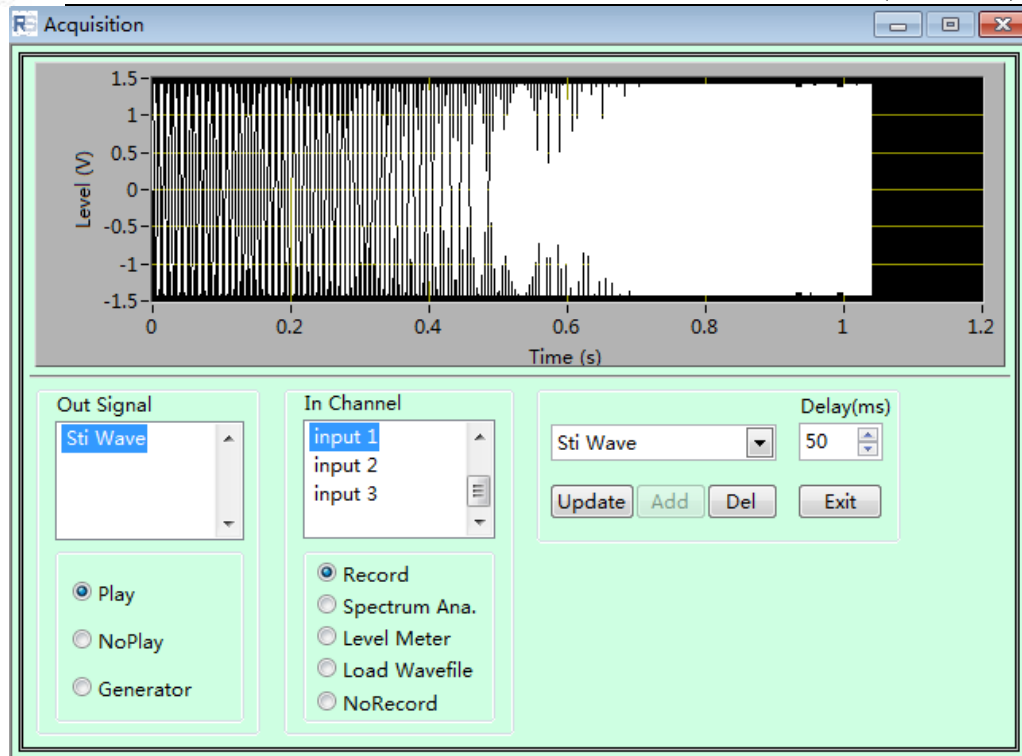
5. Test

TrustSystem mainly include these test steps: Acquisition, Analysis, Message, Com Read-Write, Loop, Command, Goto and SN. You can right click “Add” to add the step, or delete or modify the name of the step by right-clicking this step name.



5.1 Acquisition

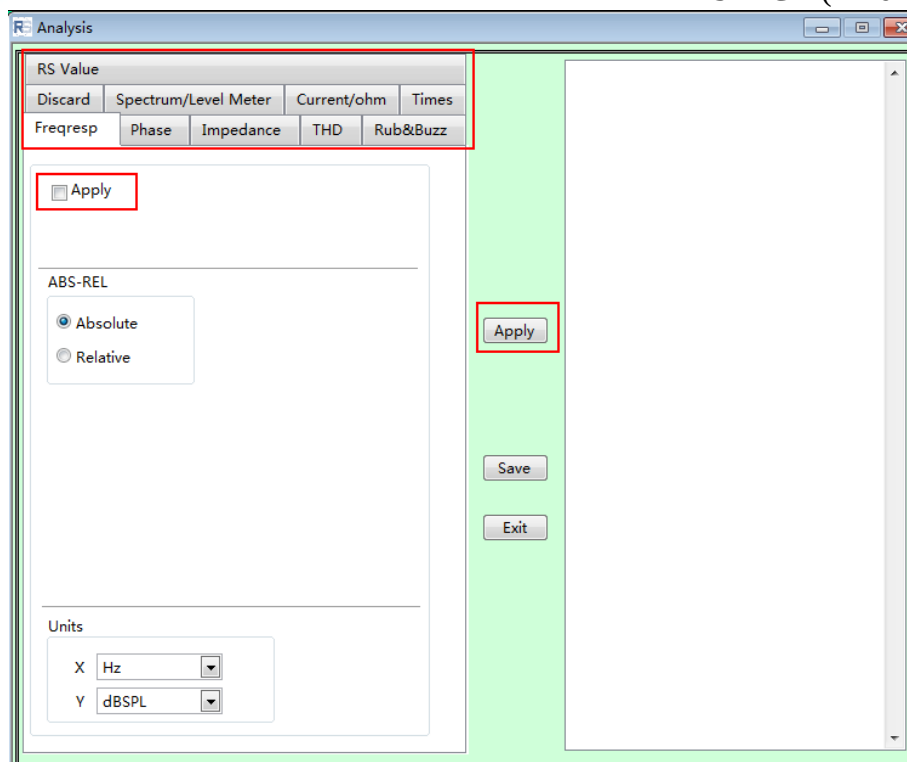
Adding the Acquisition from the Test, double clicking the Acquisition will show the interface as follows:



You can establish the out signal and input channels defined in the Stimulus and Soundcare step, Time (sec) specifies the total record time. This is typically used for measuring background noise or noise from the DUT. The minimum Record time is 50 milliseconds, there are different as the FFT, click the add button, the signal will be loaded, and the add button changes as Update button, then click save, and exit.

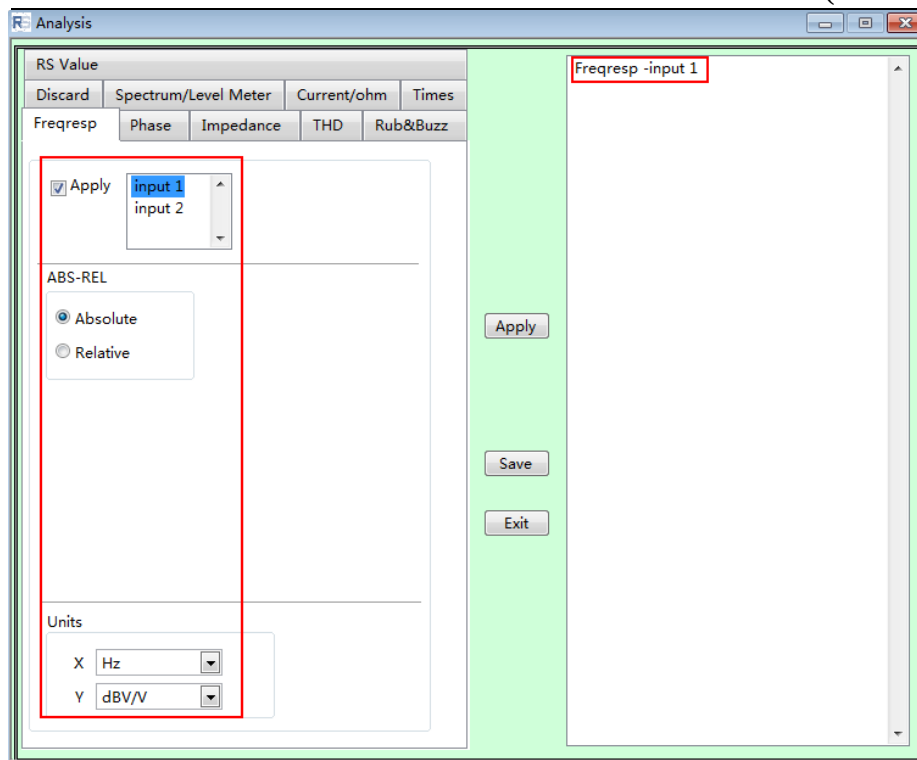
5.2 Analysis

To view and change the analysis step, please right click the mouse to add the analysis, Analysis enable you to process the signals with algorithm, you can analyze the Frequency response, Phase, Impedance, THD, Rub&Buzz normalized, Noise about the DUT. Check apply and select right channel, the parameters will be calculate automatically, when all sets have done, click done, click arrow, the testing curves and results will show the right table.



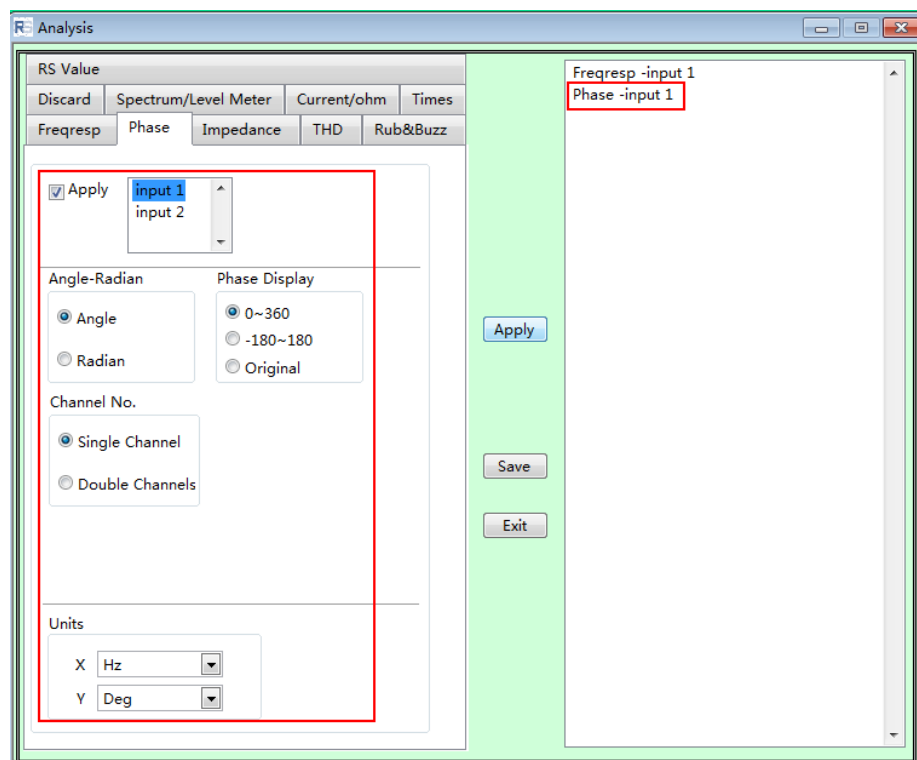
5.2.1 Frequency Response

Relative response is the response level divided by the stimulus level. The response is stimulus level independent or normalized to the input level. When increasing the input, the relative response amplitude level will not change if the system is linear, because the output level will increase by the same amount as the stimulus level. Absolute response is the response level only and stimulus level dependent, i.e.: if the stimulus increases from 1V to 2V, there will be a 6dB output level increase. The equation between dBSPL and dBP_a is: $dBSPL = dBP_a + 94$.



5.2.2 Phase

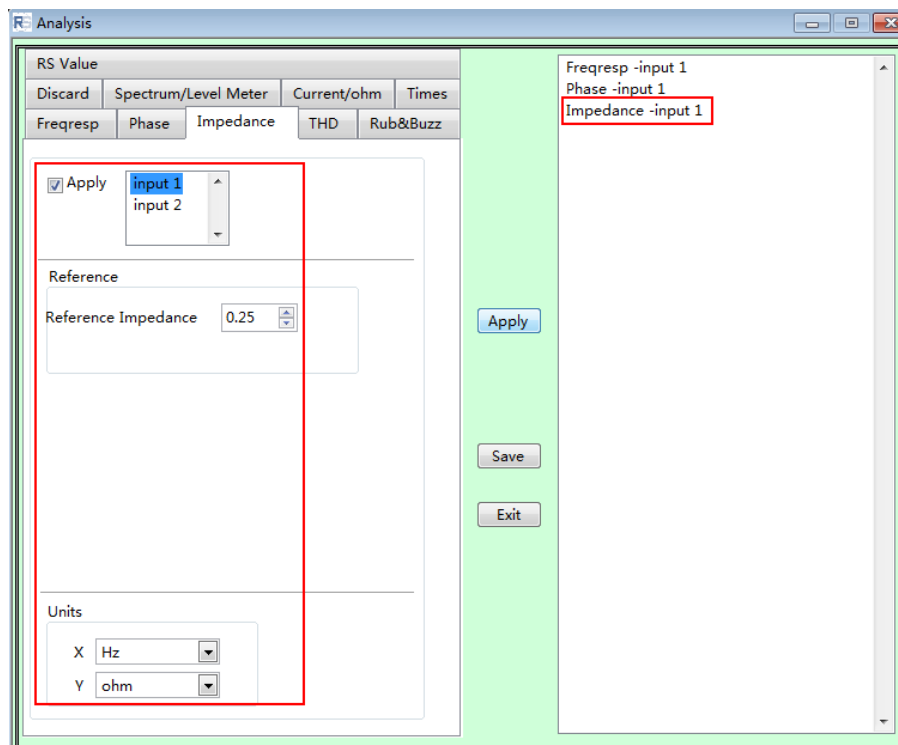
Angle-Radian: select the Phase's units, degree or Radian, the degree have 3 ranges, 0~360, -180~180 or Original, this will calculate the DUT's polarity.





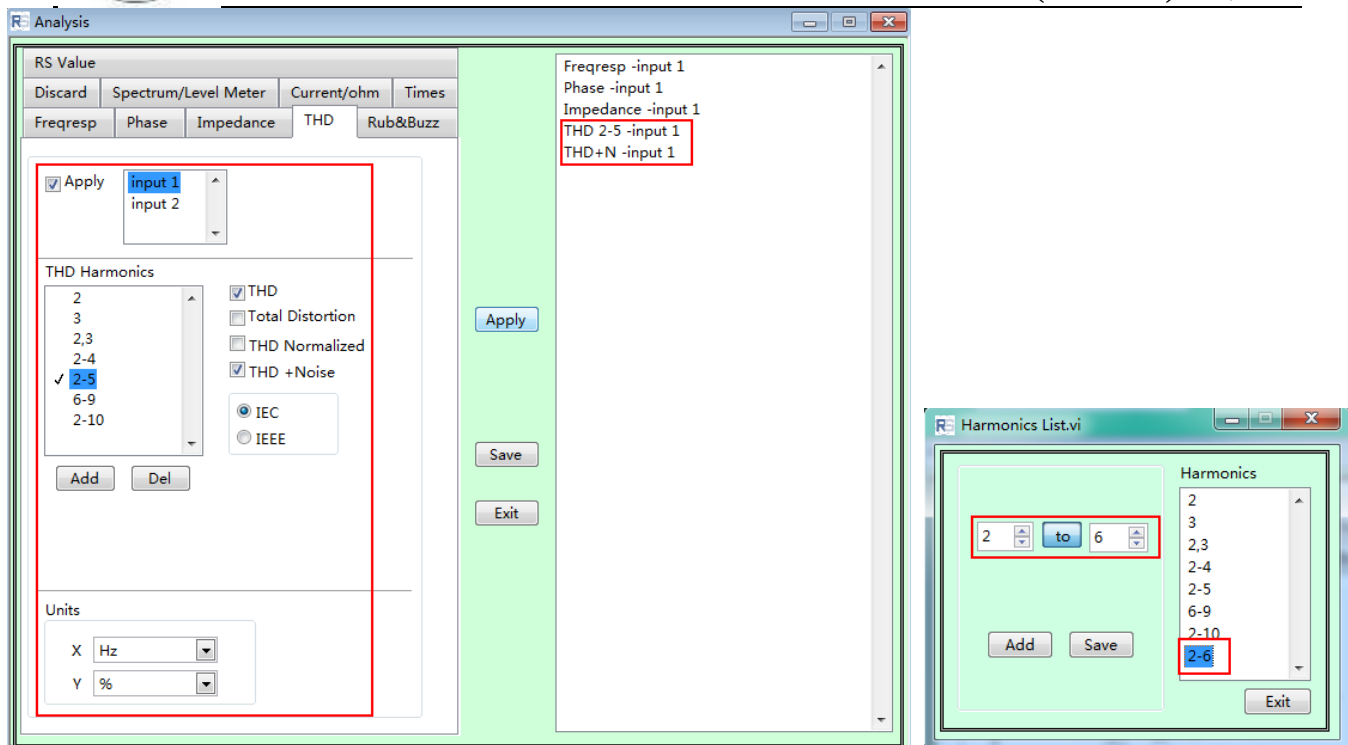
5.2.3 Impedance

To get the impedance curves, select the channel, usually select right channel, before testing the impedance, you should enter the reference impedance, the values is 0.25, the depends the values of your impedance box.



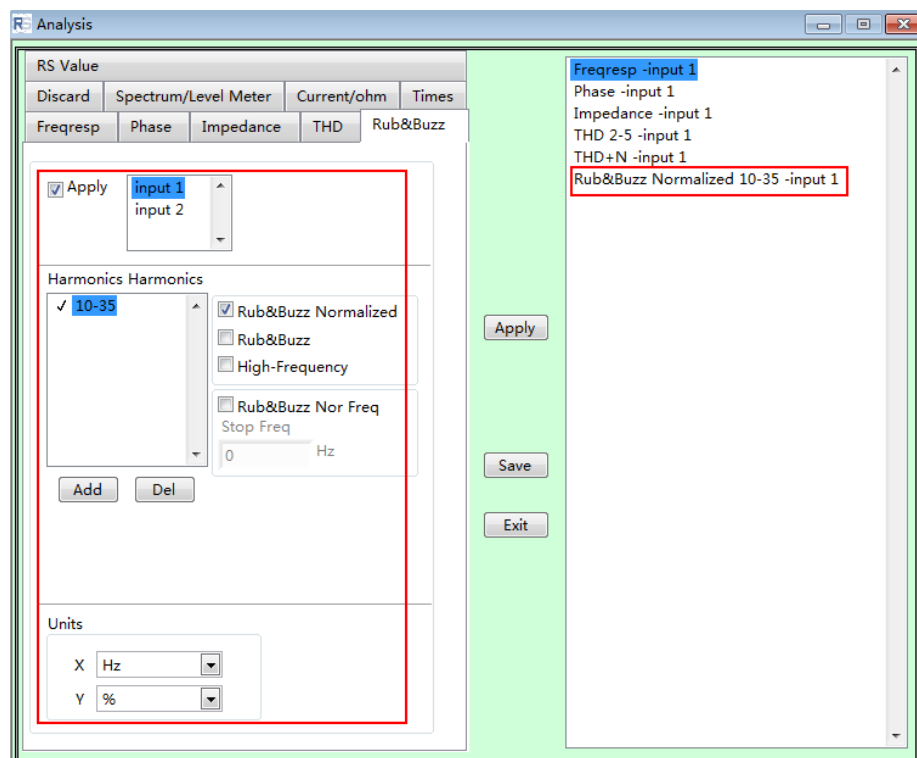
5.2.4 THD

Check the testing items (THD, THD Normalized or THD+N) and the harmonic orders by the specification, two algorithms to calculate the THD (IEC and IEEE), IEC is default, to add the harmonics orders click Add button. Add the harmonic orders, click save button, the orders will show in the Harmonics, you can add the harmonics continuously if necessary.



5.2.5 Rub&Buzz

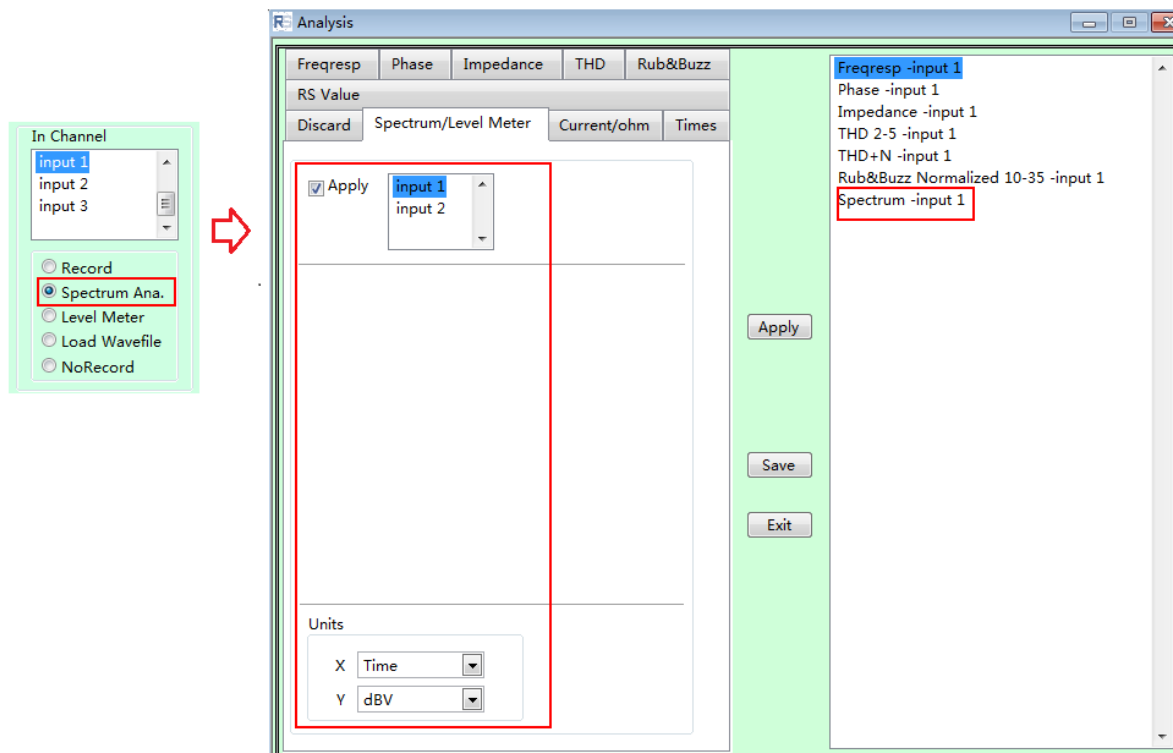
Rub&Buzz can be distinguished by people's ear, but TrustSystem can test 10~35 orders distortion and substitute for hearing, this can avoid error by hearing fatigue, so it is convenient and reliable, it also can save human resources, you can setup like THD sets, just for 10~35 harmonic orders, the result will be normalized.





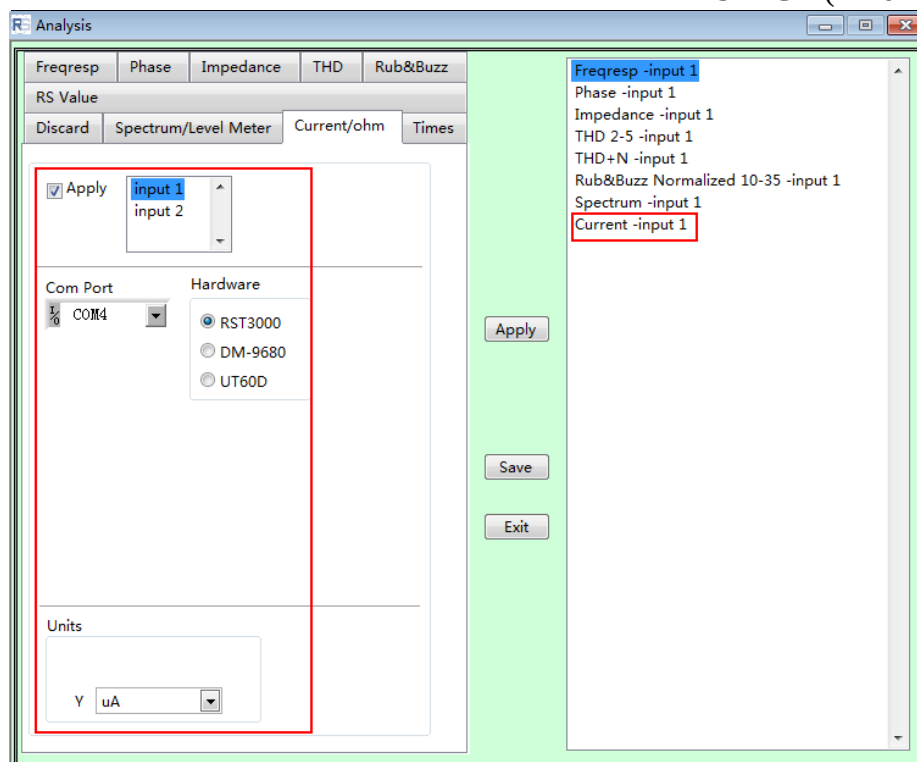
5.2.6 Spectrum

Select the appropriate channel, calculating the waveform to the spectrum, it will acquire the noise from the DUT, so we can calculate the power sum value by the noise. It's easy to get the S/N.



5.2.8 Current

TrustSystem can calculate the current from the DUT microphone when connecting the RSTech Limited hardware, such as RST3000 and DM-9680. The communication between the software and the hardware is Corm Port.

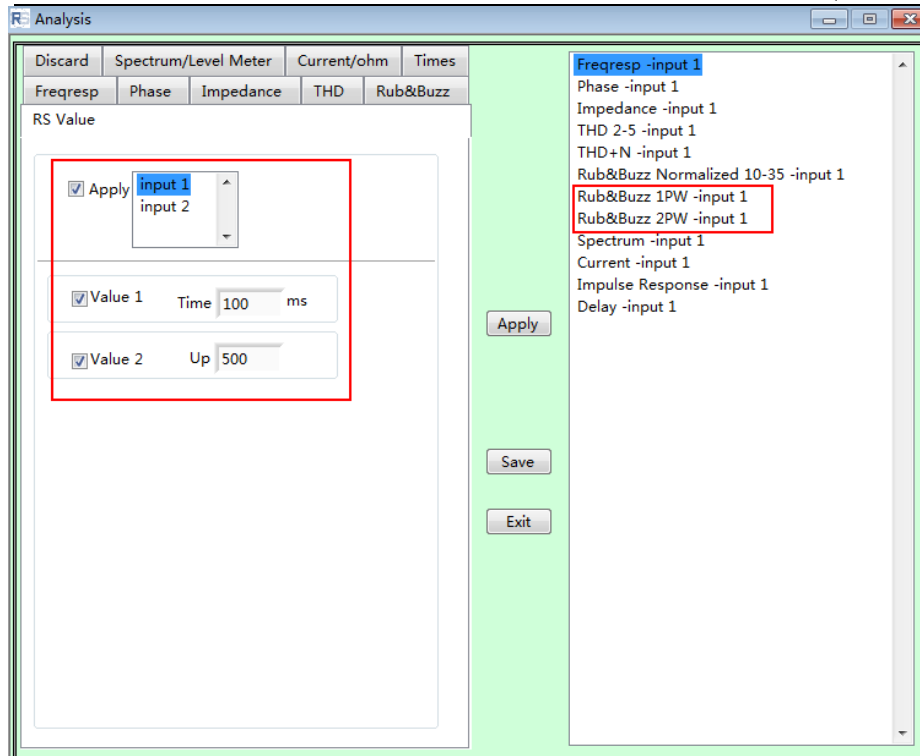


5.2.9 RS Value

Two kinds of noise calculating methods are available for user:

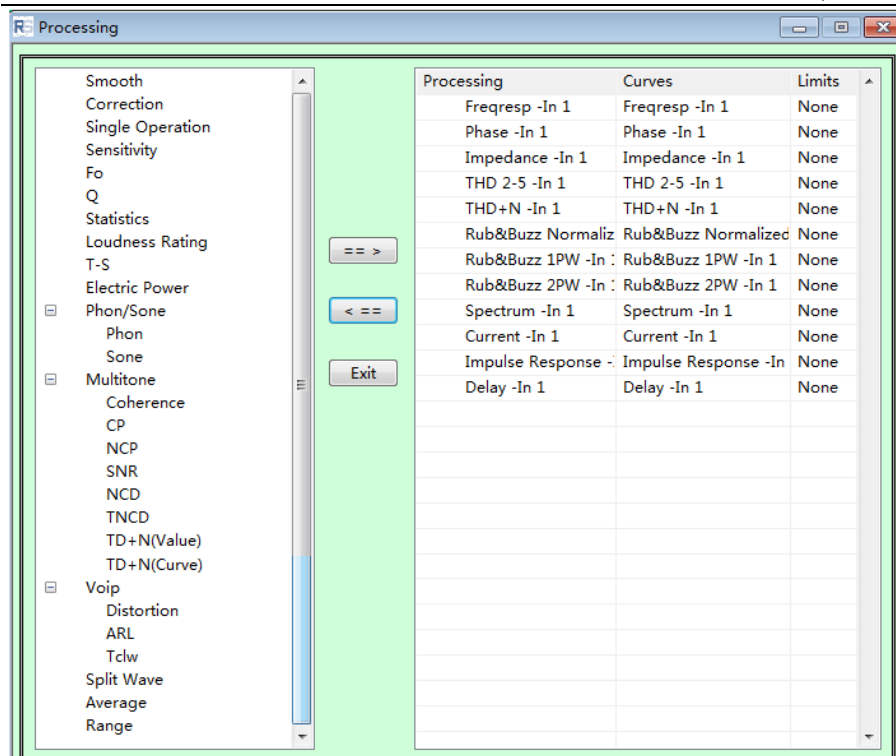
Method 1 (Value 1): Set the time length, then analyze the signals in segment, and finally obtain a dimensionless power statistic.

Method 2 (Value 2): to calculate an over-limit value by processing the signal according the set upper limit Up.



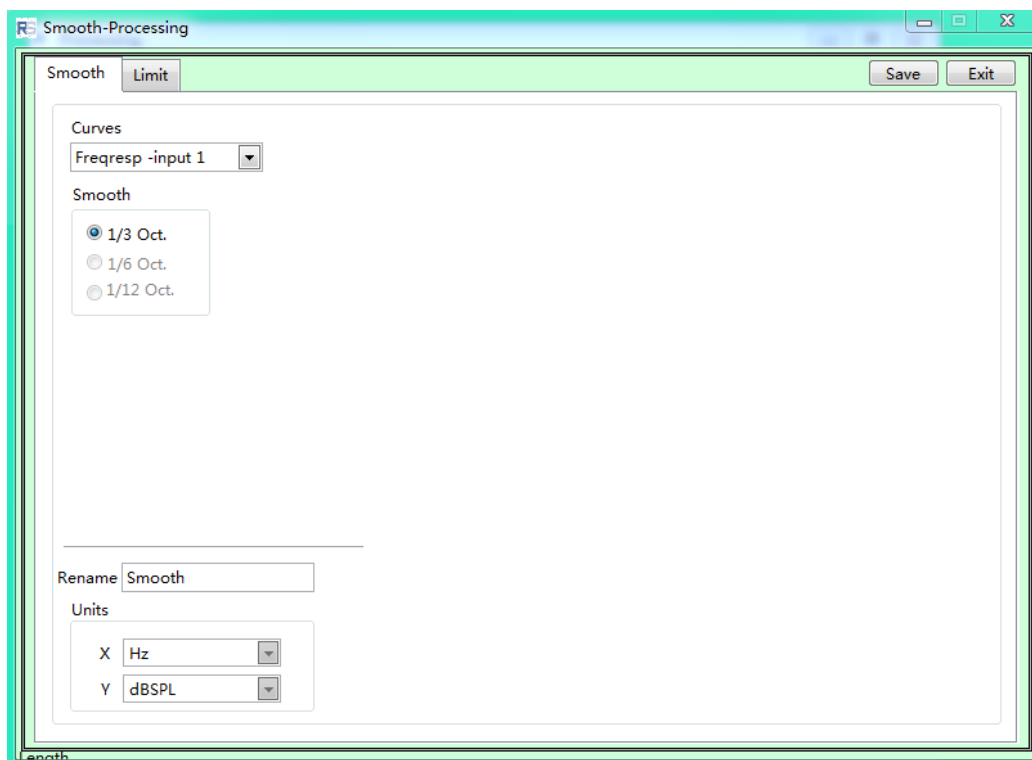
5.3 Processing

To view and change the system's post processing, double click processing from the test tree menu on the left table. Processing allows a variety of operations to be applied to measurement data and set the limits, using this step to set Pass/Fail bounds for your measured curves, and single values. A curve or value that crosses the upper or lower limit set by the step will give a Fail result, while a curve or value that lies between or equal to the set boundaries will Pass.



- Smooth

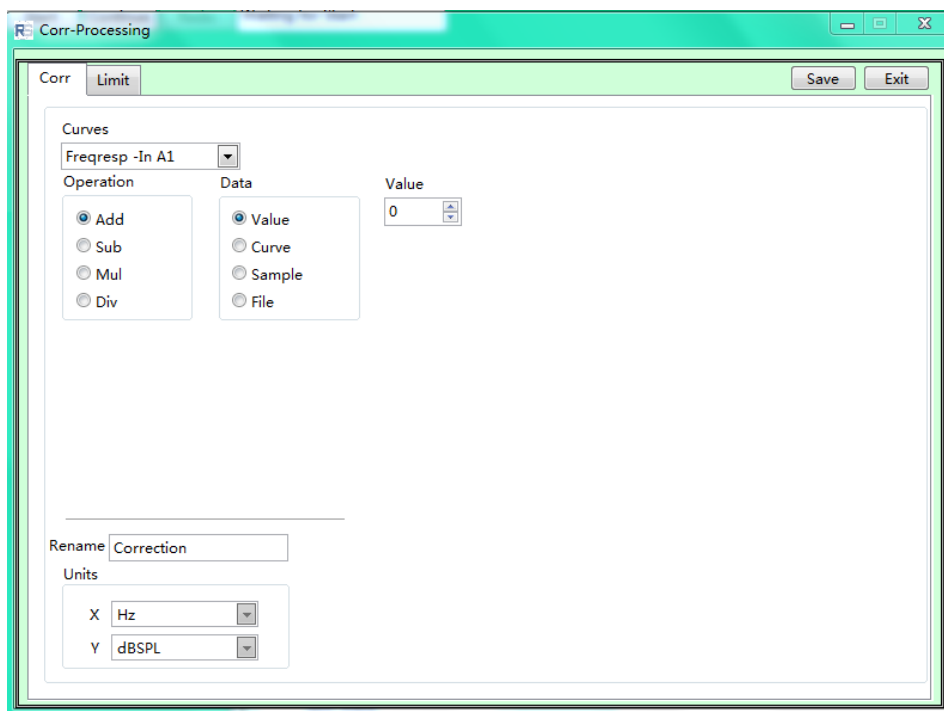
To smooth the curves that tested, select the smooth, click export in arrow button, smoothing corresponds to a running average on a frequency axis with a window of the given width (1/n octave or number of Hz). Choose the curves that you want smooth, and select the octave you smoothing, you can name the curves after smoothing, if not, smooth is the default name, then ok, and exit.





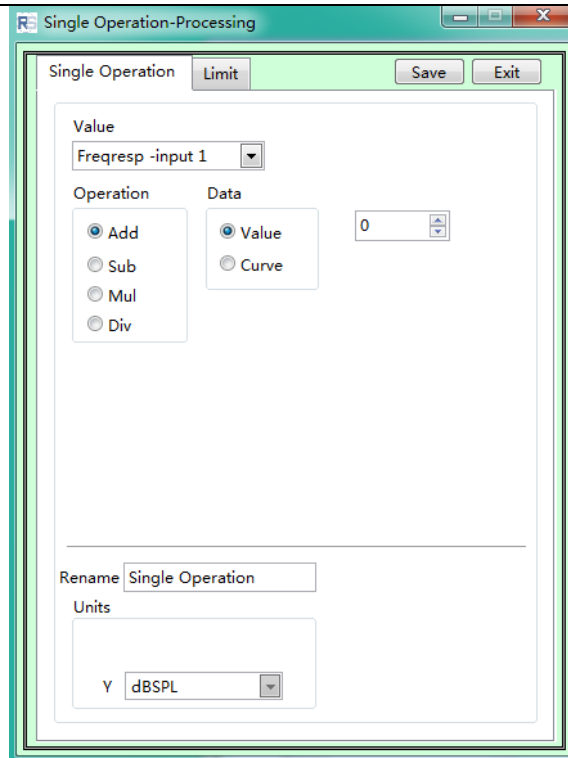
- Correction

To revise the curves that you measured, select correction, and click export in arrow button, this function will eliminate the different between two equipment's testing data, R&D and Production etc. You can choose one equipment as criteria, load the standard data of the sample product, and do the operation via TrustSystem, add/subtract or multiply/division a value/curve/sample, you can rename the curves also. To transfer the DRP (drum reference point) to ERP (ear reference point), select sample, choose the DRP-ERP data to correction.



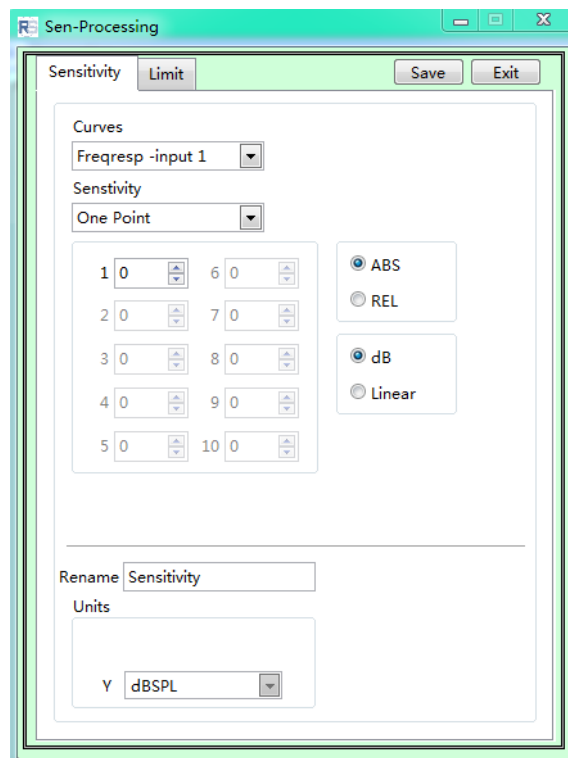
- Single operation

To correct the test single point values by using the addition or subtraction.



- Sensitivity

Average the sensitivity at different point or a range of frequencies, there are two formats for the average sensitivity: ABS or REL, ABS is the average at SPL unit, Rel is the real sensitivity, two functions can be applied: dB or Linear.

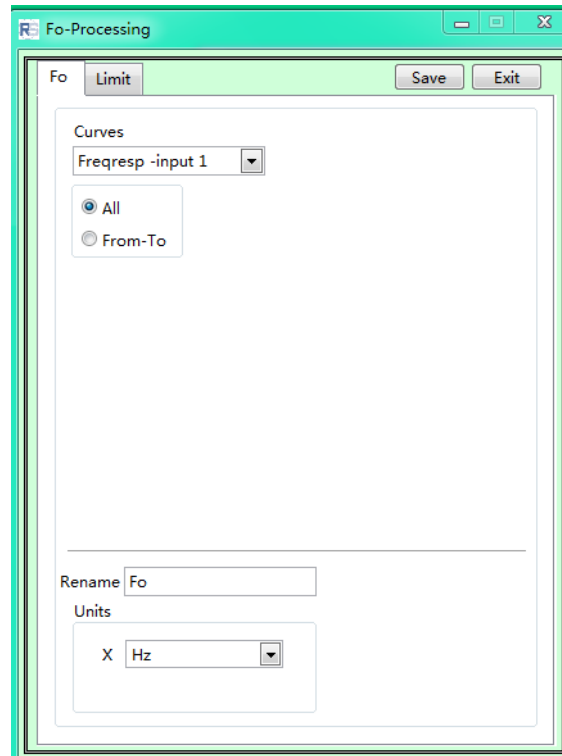


- F0 (Resonance frequency)

To get F0 and Max/Min, select the right curves, to get the value, the F0 must obtain

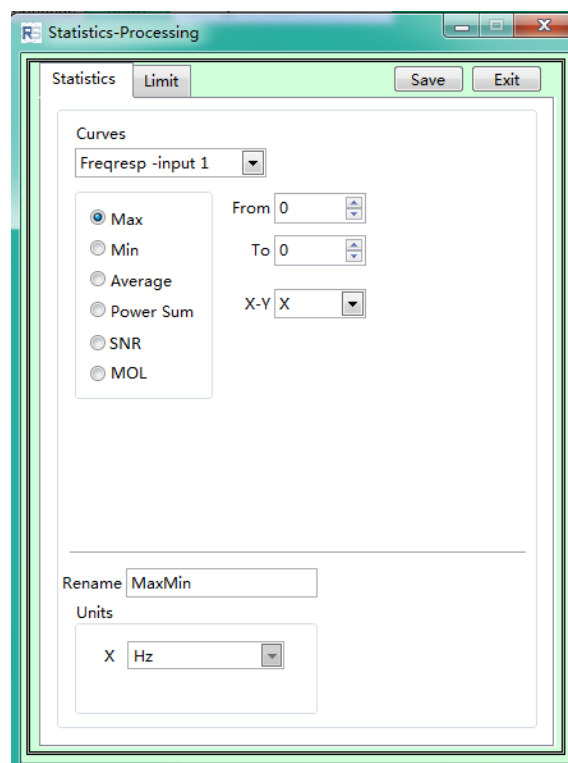


from impedance curves.



- Statistics

Find the maximum/Minimum value in the specified search range and returns Y values at that point, or average/power sum the data.

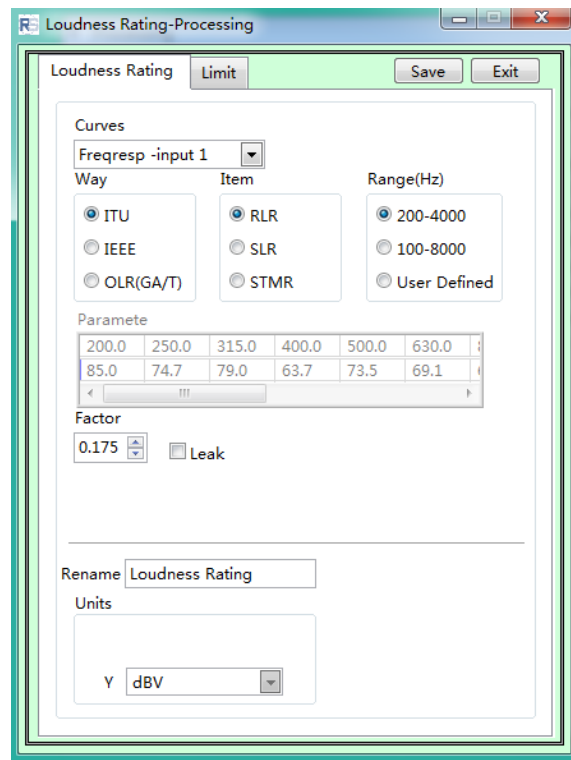


- Loudness Rating

To view and change the loudness rating, select loudness rating from processing, 3

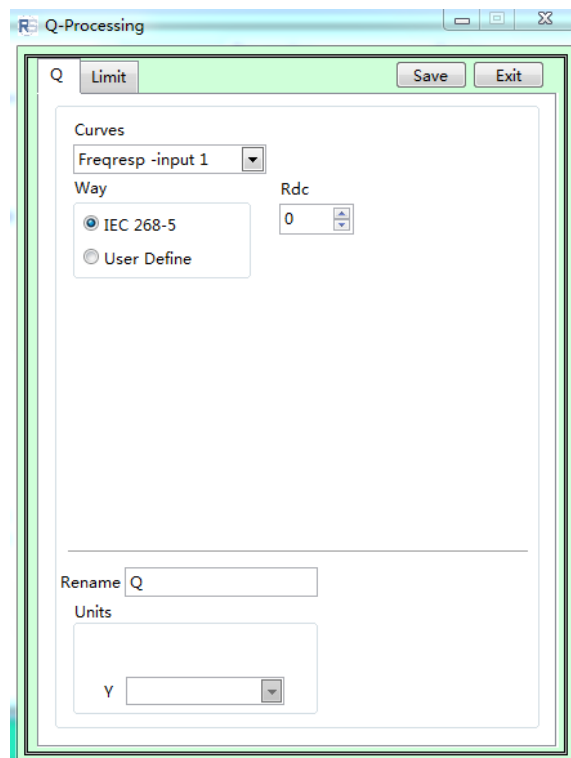


standard to choose: ITU, IEEE and OLR(GA/T); RLR, SLR and STMR can be selectable at different Frequency ranges.



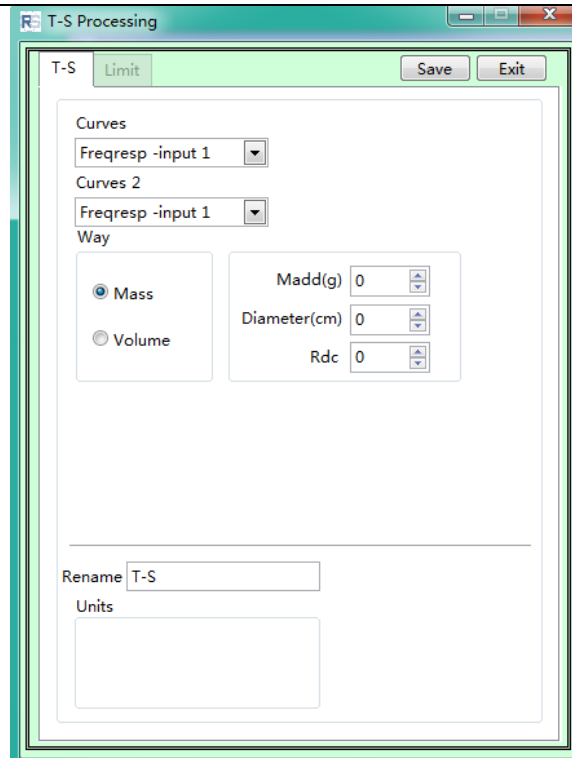
- Q value

Two standards canbe selected: IEC268-5 and User define.



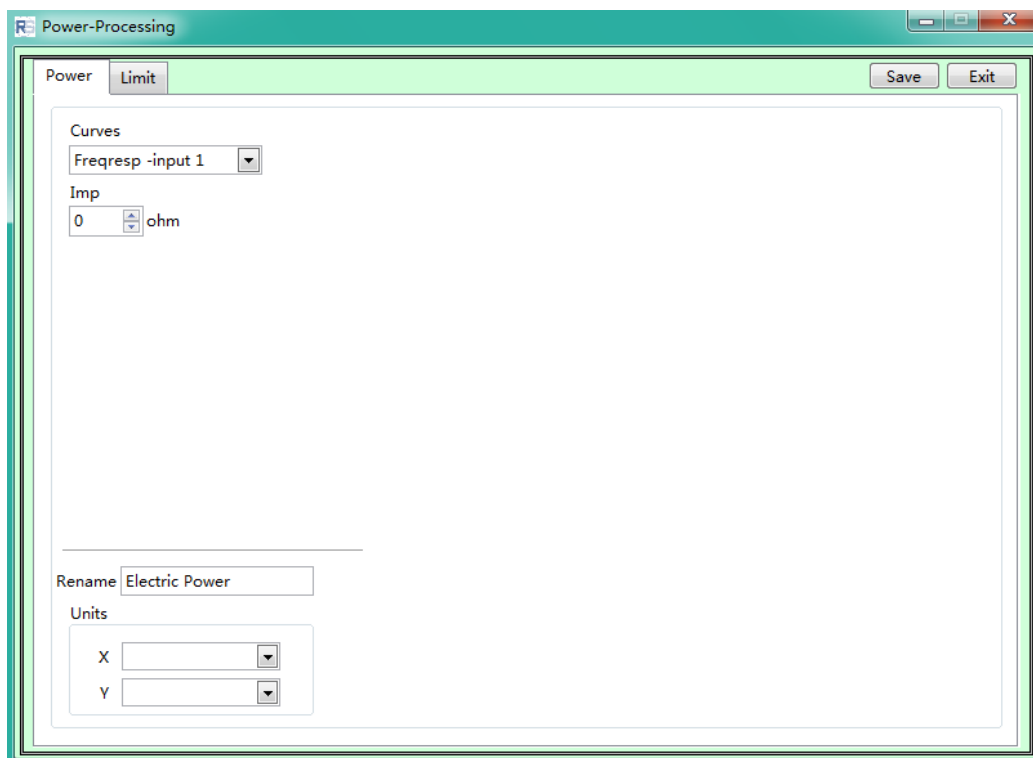
- T-S

TrustSystem can calculate the T-S parameters, enter your parameters by your DUT.



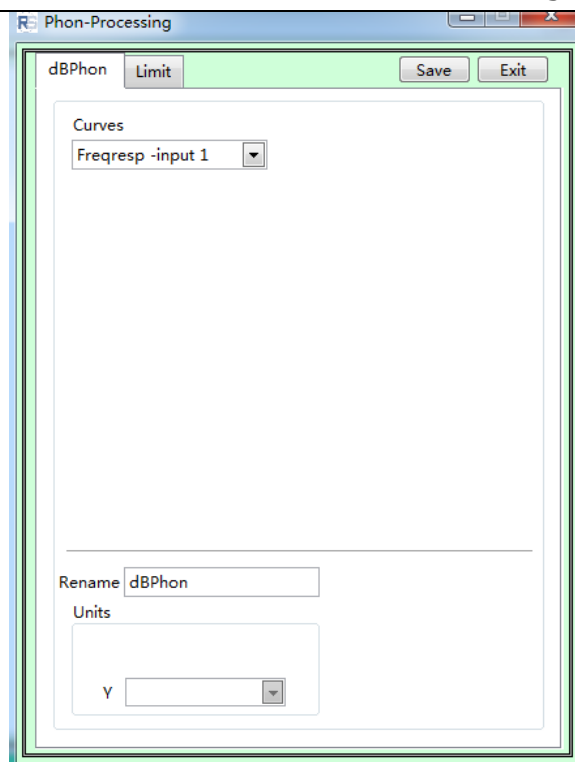
- Electric Power

Select a frequency curve and input the impedance in this dialog to get the corresponding electric curve.



- Phon / Sone

The loudness calculation conforms to ISO 532 international standard.

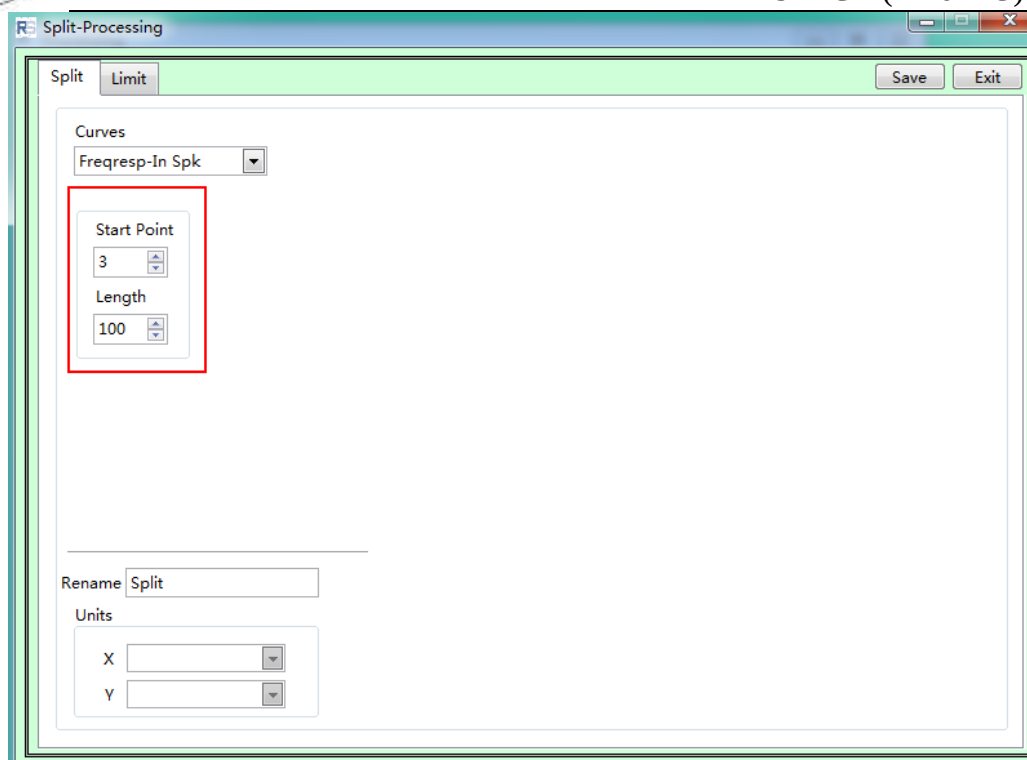


- Multitone / Voip

Multi-tone is an ensemble of tones regularly spaced in frequency. The amplitudes are equal and to minimize the crest factor, the phase of the tones is random. To calculate the coherence, CP, NCP, SNR, NCD, THCD OR Voip, please view the processing function, it includes different standards TIA-810/920, ITU P.311 etc.

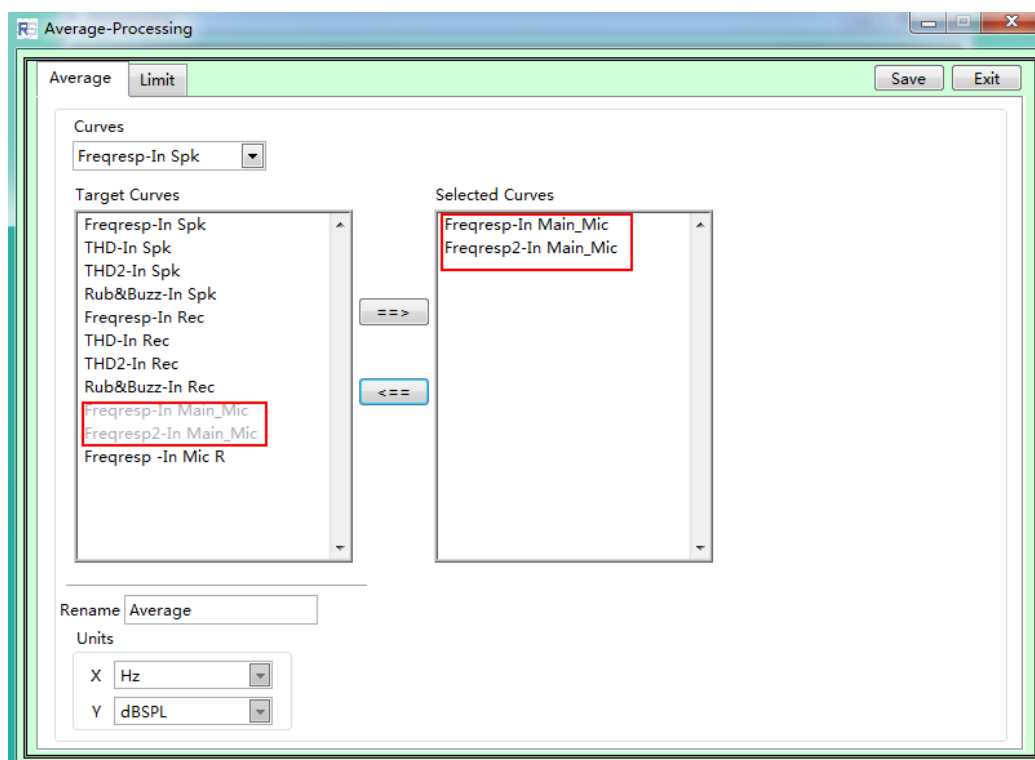
- Split Wave

To split the test curve into sections by entering the Start Point and the Length of the data segment.



- Average

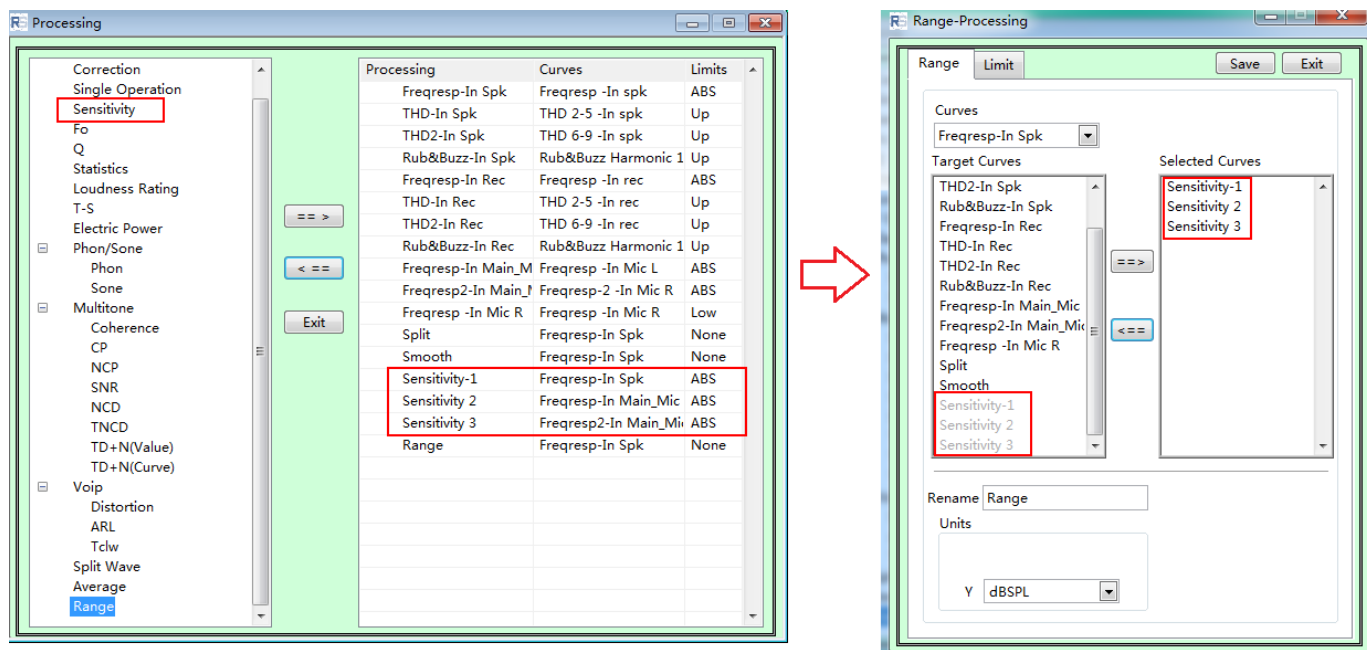
To calculate the average of multiple frequency response curves. Select multiple frequency response curves in Target Curves, add them to the right list. Click to save it and exit. After the analysis is completed, an average frequency response curve can be obtained.



- Range



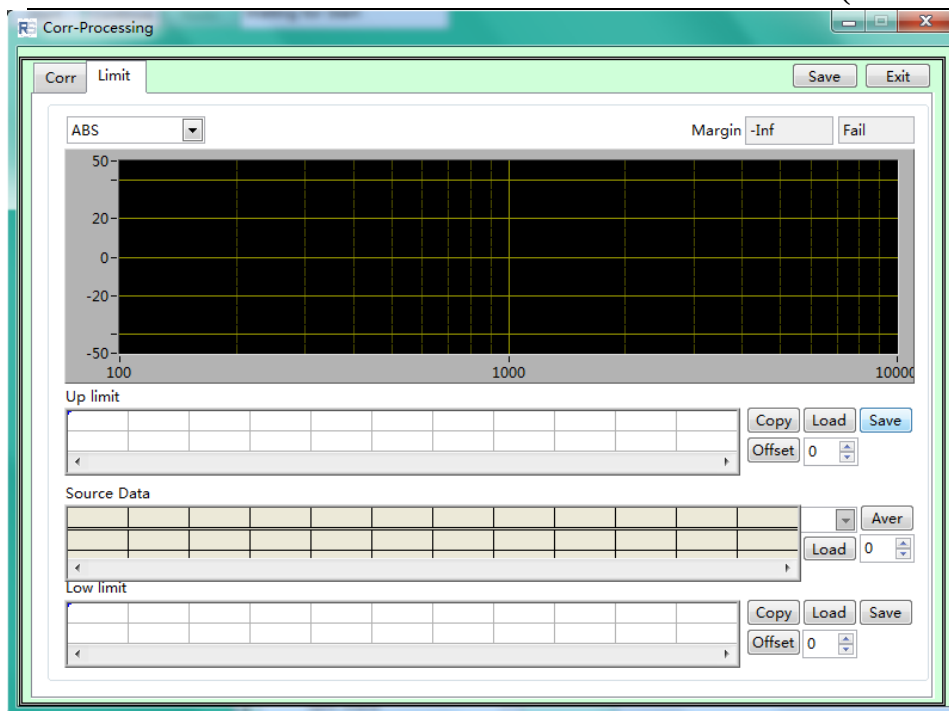
To calculate the range of multiple sensitivity test results. Select multiple sensitivity test results in the Target Curves of the Range window, add them to the right list, click to save it and exit.



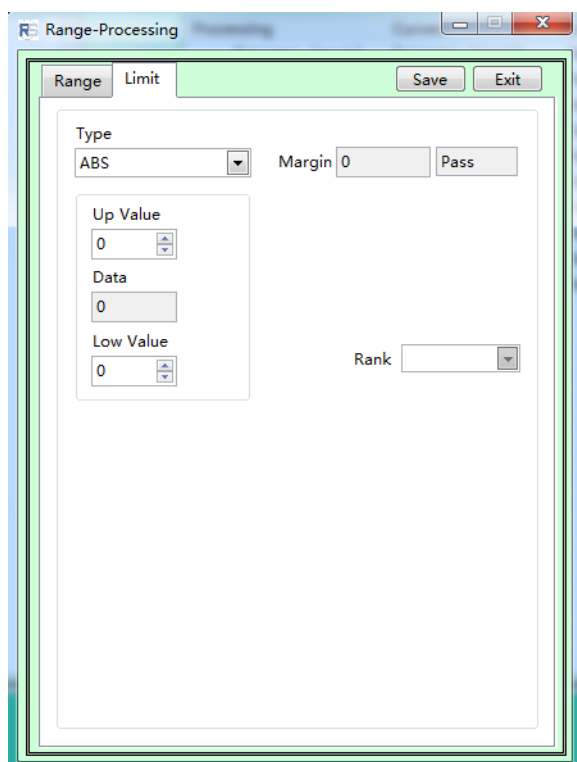
● Limit

To view and change the system's test limits, select processing from the test menu on the main TrustSystem menu. There are many function, each can set Pass/Fail bounds for your measured curves, and single values. A curve or value crossed the upper or lower limit set by the step will give a Fail result, while a curve that lies between or equal to the set boundaries will pass.

Curves: there are many different ways of applying limits to measurement data. Options include: 1)Rel, setting absolute limits; 2)ABS. allowing data becomes zero at 1kHz; 3)Up limit, up limit only; 4)No, no limits. Data from current curves can be used as a limit curve, i.e. ± 3 dB of running curve, when the limits set over, enter the limits name in the Rank, and click OK, and exit.



Value: you can select limit to make limits effective. Options include: 1) None, no limits; 2) ABS: setting absolute limits; 3) UP, up limits only; 4) Low, low limits only. When the limits set over, enter the limits name in the rank, and click OK, and exit.

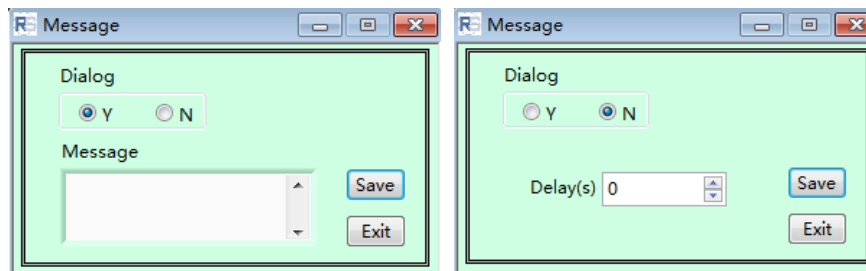


5.4 Message

The message can be added in the project, to provide the message to the operator: 1) Y, display the messages on the computer screen for the operator to read; 2) N, control length of

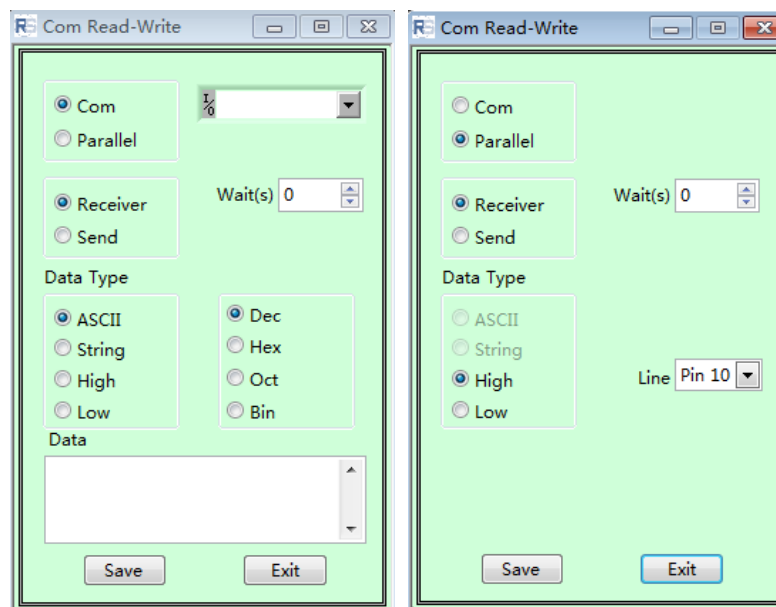


time (in milliseconds) the message step waits before finishing. This can be used to ensure commands or devices settle before continuing to measure.



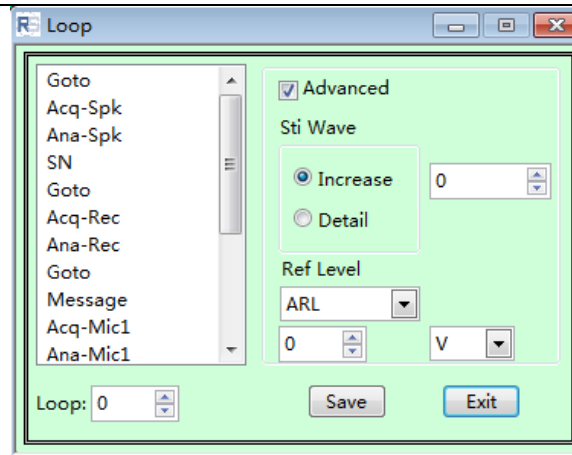
5.5 Com Read-Write

Communicate devices with PC's RS-232 and IEEE-488 interface, it have two ways to communicate your devices: receive and send with different formats. String: input the string to communicate, ASCLL: have four formats: Dec, Hex, Oct, Bin.



5.6 Loop

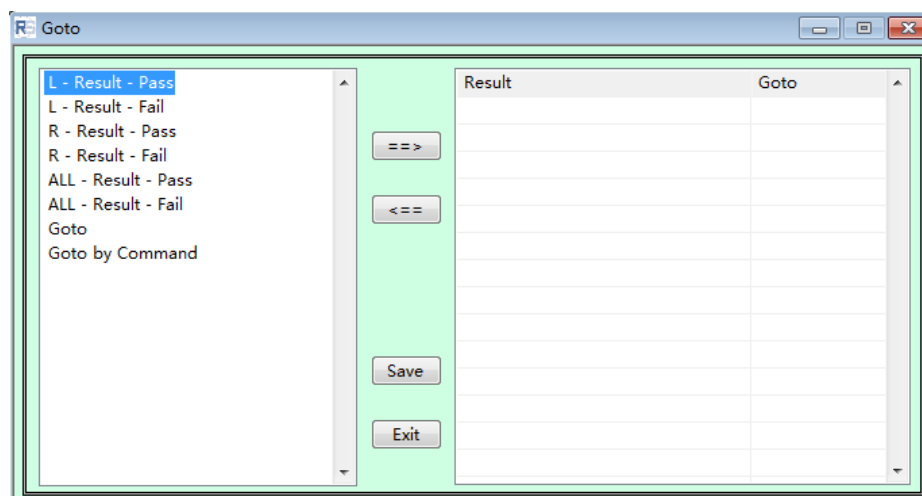
TrustSystem can do some steps continuously by Loop function. Select the steps for loop, set the stimulus and times for the testing. Detail function can be edited the precise values to loop.



5.7 Go to

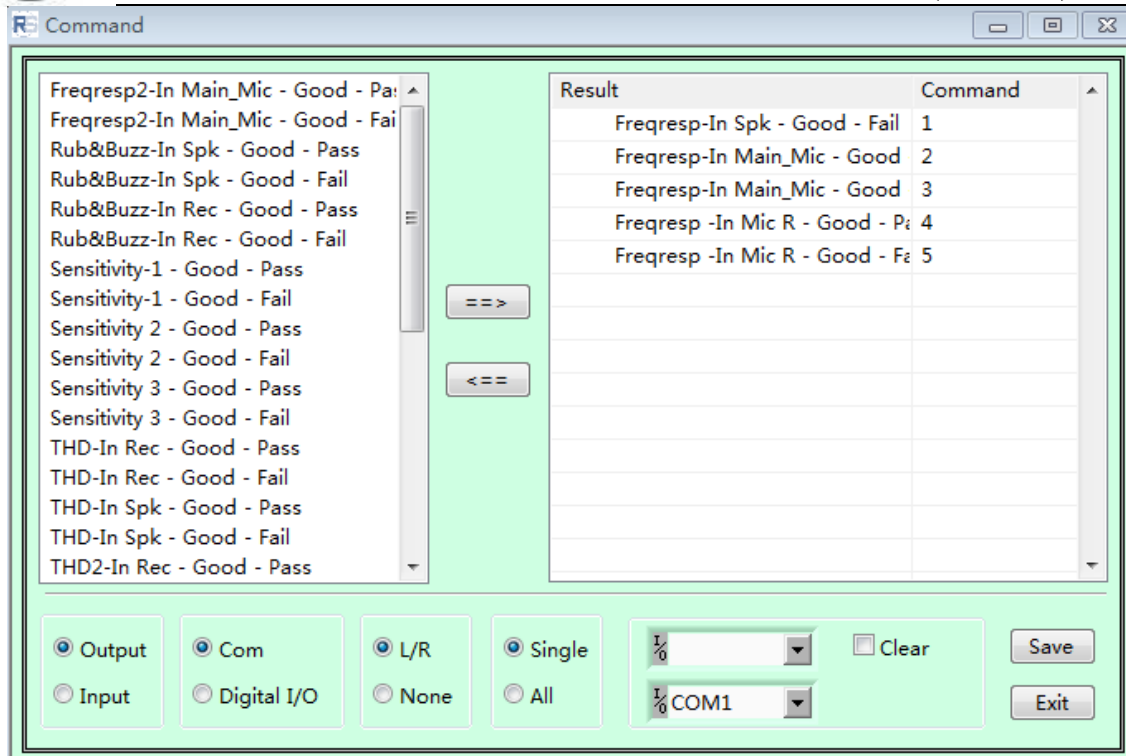
When the test meets certain conditions (Pass or Fail), you can select this item to go to the next step. Goto can be used in conjunction with Loop instructions, in which the number of loops and Loop steps can be set.

Note: you need to set Goto to determine the jump, then set the Loop number and the Loop test step in the Loop.



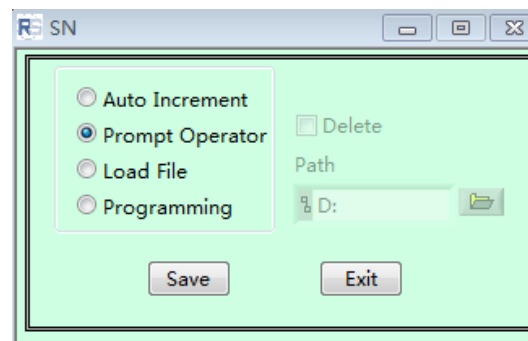
5.8 Command

TrustSystem have two ways to communicate the external software: Com and Digital I/O (NI 6503), it is designed for different ranges for DUT.



5.9 SN

Serial number control: the optional operations include Auto increase, Prompt Operator, loading the saved files, etc.



5.10 Save

To view and change the system's Save, select Save from the test menu on the main TrustSystem menu. There are two tabs to choose: save item and file setup,

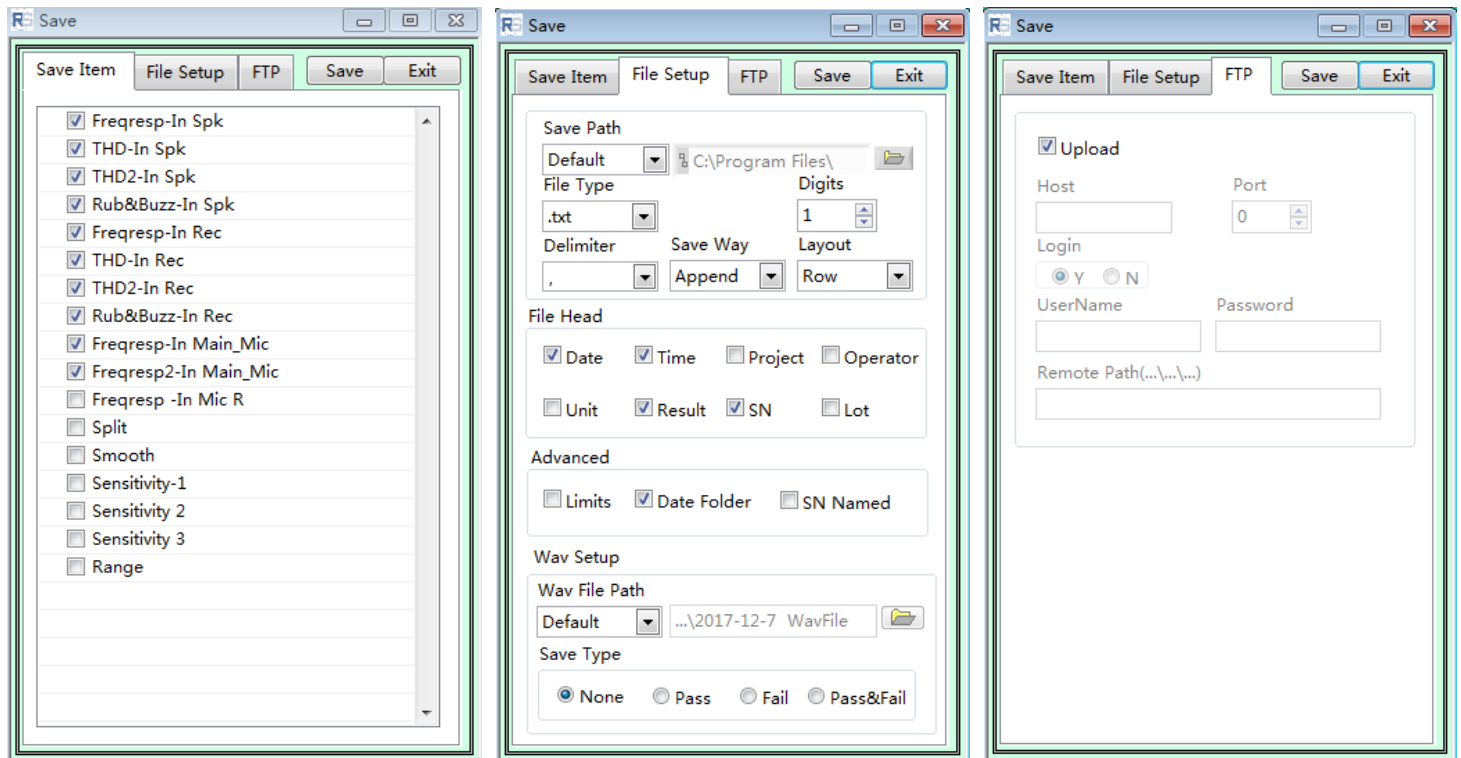
Save Item: check the curves or values to save.

File Setup: 1) Save Path, select default or user (user type, user can define their save path); 2) File Type, select tab and comma type; 3) Delimiter, comma and Tab; 4) Save way, new(create a new file), append(add the data at same file), overwrite(overwrite the data); 5) Save type, Row or Colum, means the data format.

FTP (File Transfer Protocol): check Upload to upload the selected result file to the shared



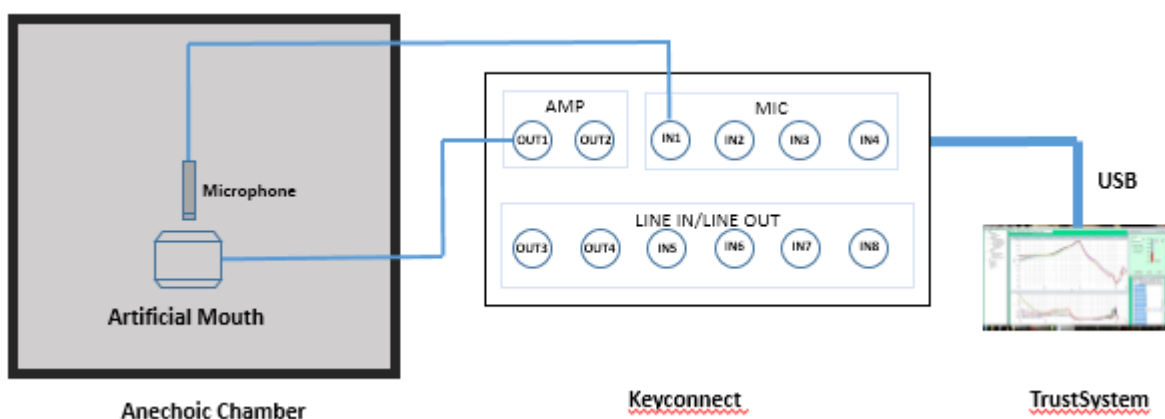
folder.



6. Test Case

6.1 Microphone Test

- Hardware Configuration: Keyconnect (Multi-channel audio test system, which includes power amplifier and microphone power adapter), AM3000 Artificial mouth which conforms to the Standards IEEE269 and ITU-T Rec.P51, RST4000 Prepolarized 1/2 “capacitive high precision free field microphone, the DUT microphone.
- Connect schematic diagram



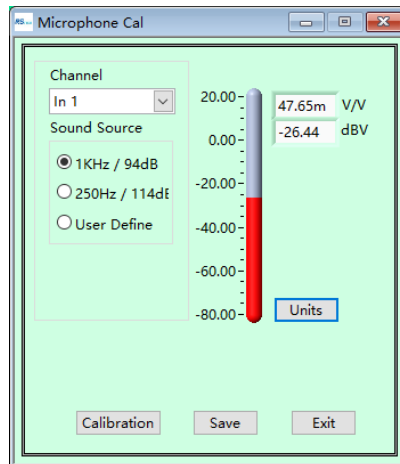
- Test Methods



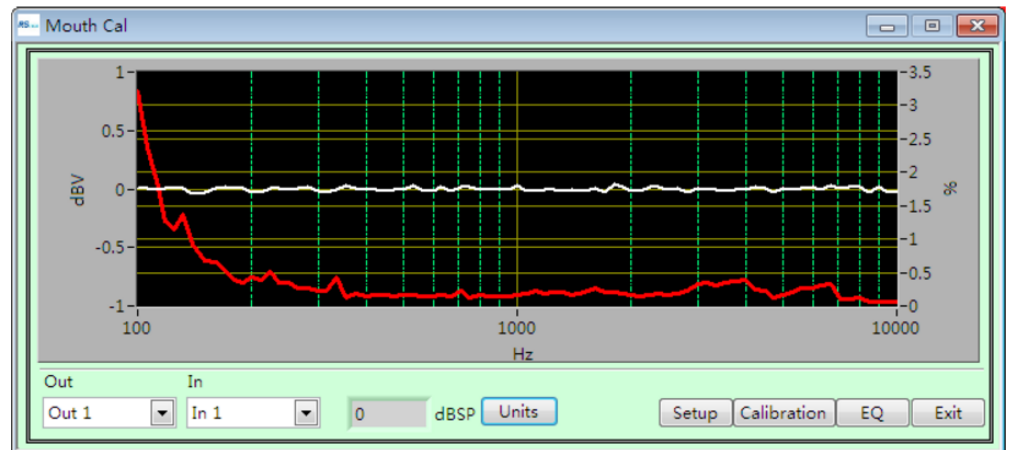
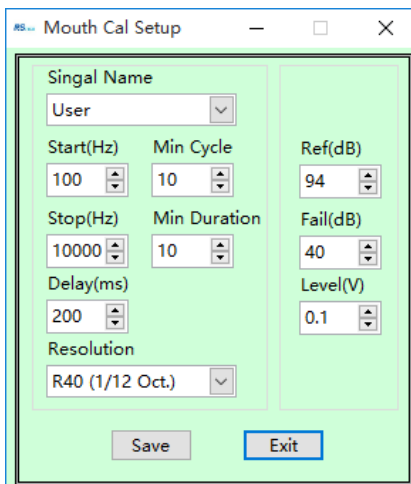
Step 1: Connect the RST4000 to the MIC input channel of Keyconnect; Connect the AM3000 to the amplifier output channel of Keyconnect;

Step 2: Calibrate the RST4000 and save the sensitivity;

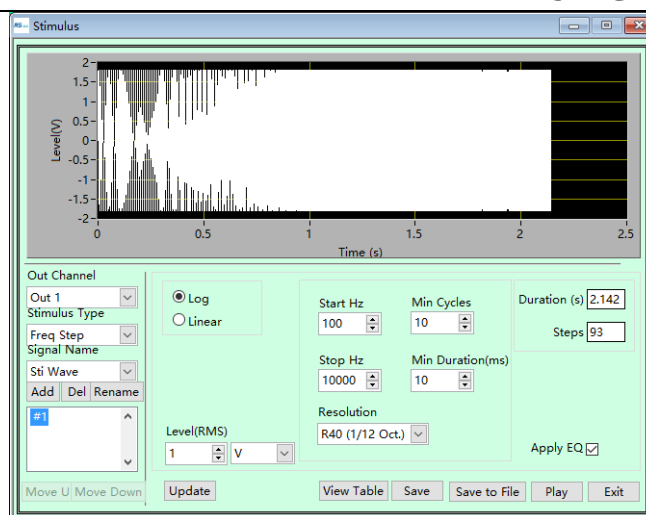
Step 3: Put the RST4000 on AM3000 and calibrate the artificial mouth;



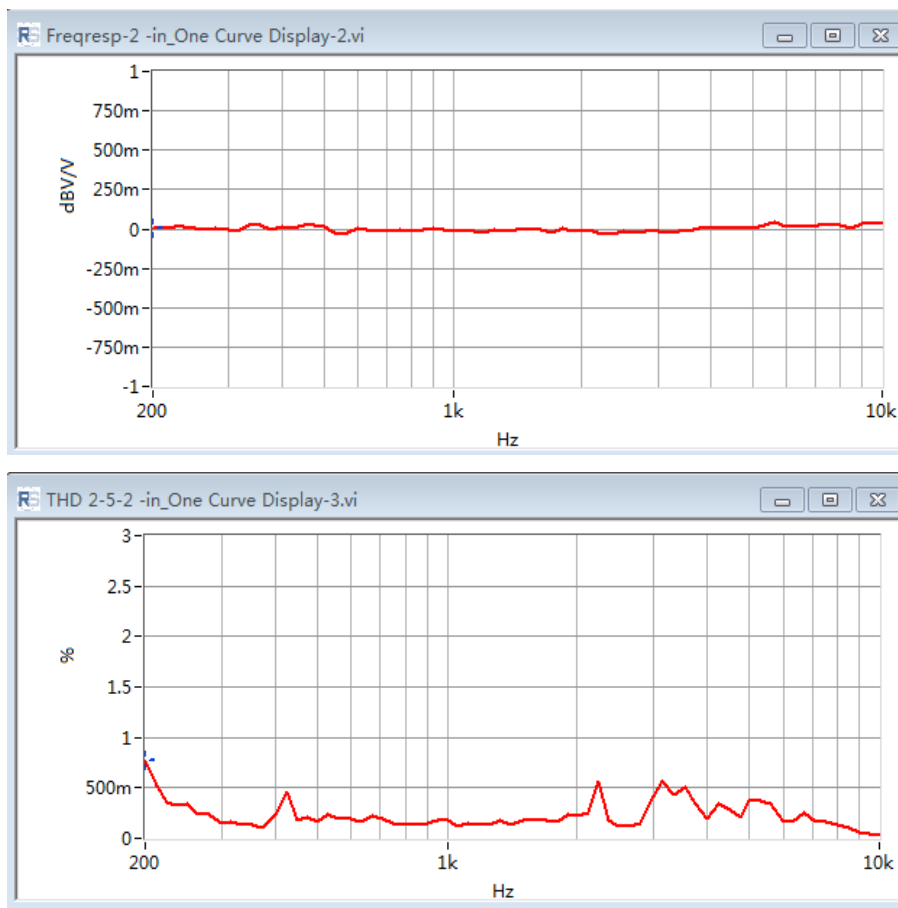
Step 4: click the Mouth Cal, and set the relevant parameters in Setup dialog (Frequency band width, 1/12OCT, 94 dB Ref); click Calibration, and then click EQ when the value of dBSP is near 94dB. Click Continue until there is a sufficiently flat frequency curve;



Step 5: Put the DUT microphone on the AM3000, set up the parameters in Stimulus as follows; calibrate the DUT microphone and save record the sensitivity;



Step 6: Use the TrustSystem to output sweep signal and analysis the frequency response, distortion, etc.

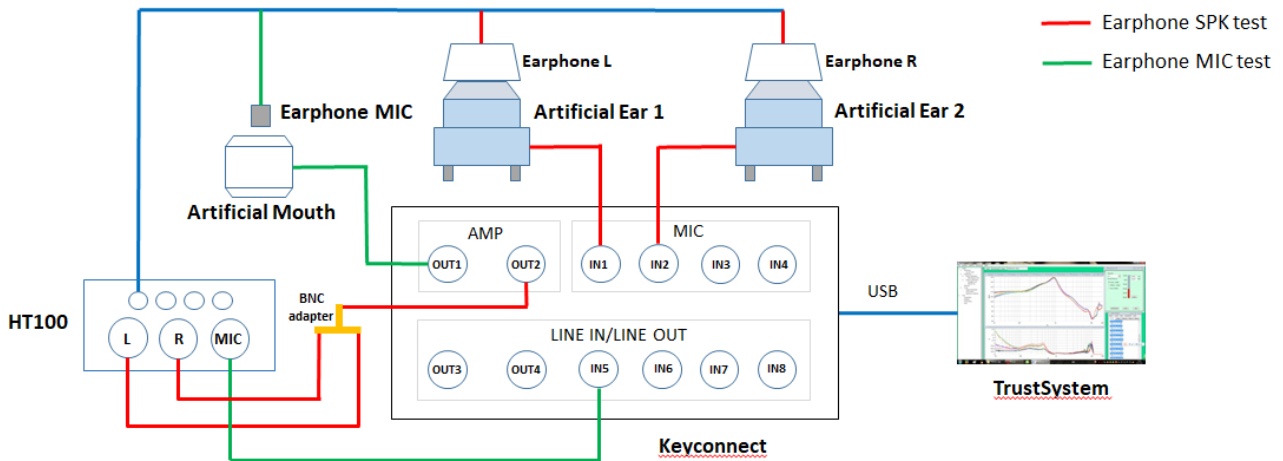


6.2 Earphone Test

- **Hardware Configuration:** Keyconnect (Multi-channel audio test system, which includes power amplifier and microphone power adapter), HD400(Headphone Test Jig, which includes 1pcs test jig, 1pcs Artificial Mouth AM3000 and 2pcs Artificial Ear), HT100 Earphone Test Interface, DUT Earphone (In-ear).



- Connect schematic diagram



- Test Methods

Step 1: Put the DUT earphone on the HD400 bracket;

Step 2: Plug the earphone into the HT100, and connect the HT100's L/R channel to the amplifier output channel of Keyconnect; connect the HT100's MIC channel to the LINE IN channel of Keyconnect;

Step 3: Connect the artificial mouth to the amplifier output channel of Keyconnect;

Step 4: Connect the artificial ear to the MIC

Input channel of Keyconnect;

Step 5: Use the TrustSystem to output sweep signal and analysis frequency response, distortion, etc.

